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journal homepage: www.elsevier.com/locate/jfecDo the rich gamble in the stock market? Low risk anomalies and wealthy households[☆]Turan G. Bali^{a,*}, A. Doruk Gunaydin^b, Thomas Jansson^c, Yigitcan Karabulut^{d,e}^a McDonough School of Business, Georgetown University, United States^b Sabanci Business School, Sabanci University, Turkey^c Sveriges Riksbank, Sweden^d Frankfurt School of Finance & Management, Germany^e CEPR, United Kingdom

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ABSTRACT

Contrary to the theoretical principle that higher risk is compensated with higher expected return, the literature shows that low-risk stocks outperform high-risk stocks. Using a large-scale household dataset, we provide an explanation for this puzzling result that the anomalous negative risk-return relation is only confined to those stocks predominantly held by rich households, whereas the anomaly disappears for stocks held by non-rich households and institutional investors. We find that social status concern of rich households and the induced lottery preference explain wealthy investors' demand for high-risk stocks, leading to overpricing and low future returns for such stocks.

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1. Introduction

Levy (1978) and Merton (1987) provide theoretical evidence that if investors do not hold a large number of assets in their portfolios, firm-specific (idiosyncratic) risk affects equilibrium asset prices and commands a positive risk premium. Contradicting established theoretical models, Ang et al. (2006) – the most widely cited study on idiosyncratic risk – present empirical evidence of a *negative* cross-sectional relation between idiosyncratic volatility and future equity returns. This result is highly inconsistent with theoretical predictions and, thus, considered a puzzle.

A rich subsequent literature has proposed several explanations for the anomalous negative risk-return relation among high-volatility stocks – also known as the idiosyncratic volatility (IVOL) puzzle.¹ One common finding across these studies is that individual investors, who are assumed to have non-standard preferences and are financially less sophisticated, play an important role in the existence of the IVOL puzzle. However, this explanation does not align well with the evidence that the direct stock ownership rates across the population are particularly low around the world. For example, in the U.S. where the overall stock market participation rate is one of the highest in the world, only 14% of the population directly invest in the stock market.² Hence, an essential question is how can so few individual investors affect equilibrium asset prices and contribute to the asset pricing puzzles, as argued by the extensive literature in asset pricing?

We provide empirical evidence that the highly skewed distribution of household wealth offers a potential explanation. In principle, only a very small subset of individual investors can affect asset prices if most of the (stock) wealth is concentrated in the hands of these investors. Consistent with this notion, we show that the IVOL puzzle is primarily driven by the demand of rich individual investors who are in the top 10% of the wealth distribution and hold most of the stock wealth in the economy. Interestingly, the low-risk anomaly disappears for stocks widely held by non-rich households or institutional investors. Using household level information on the stock investments of an entire population, we also scrutinize the economic mechanisms underlying the demand of rich households for high-volatility stocks. We document that social status concerns of rich households and the induced lottery preferences explain wealthy investors' demand for high-risk stocks, leading to overpricing and low future returns for such stocks (Becker et al., 2000; Roussanov, 2010).

We begin our empirical analysis by studying the stock investment behavior of individual investors at the household level with a particular focus on whether rich house-

holds tend to tilt their portfolios towards stocks with higher idiosyncratic volatility. Our dataset, which provides highly accurate information on household assets, disaggregated at the security level, of every individual residing in Sweden, enables us to identify the relative position of each household in the entire wealth distribution, and to directly observe and analyze their stock investment behavior over the 1999–2007 period.

Our household level portfolio analysis reveals an interesting pattern. When we consider the entire universe of stock investors, we document a significantly negative relation between household wealth and household portfolio exposure to idiosyncratic risk. In fact, this finding is in line with Kumar (2009) who finds a negative correlation between individual wealth and weight of lottery-type stocks in the equity portfolio of retail investors. Interestingly, when we divide the population into two subsamples by their rank in the wealth distribution, i.e., top 10% vs. bottom 90%, we find a significantly positive effect of wealth on the portfolio IVOL of richer households but a negative effect for households in the bottom 90% of the wealth distribution. These results are robust to controlling for a rich set of household and portfolio characteristics as well as a battery of fixed effects for household, region of residence, and industry of employment, respectively. Hence, it appears that increases in wealth tend to exhibit opposing effects on the risk-taking behavior of households who are in different social classes as measured by their relative position in the wealth distribution. To refine our understanding on the non-linear effects of wealth on household risk taking, we then focus on changes in the relative position of households in the wealth distribution, and show that households who move up to the top of the wealth distribution (i.e., top 10, top 5, or top 1%) significantly increase their stock portfolio exposure to IVOL. Interestingly, the documented effect increases with the relative wealth rank of households in a non-linear fashion. Taken together, these results are consistent with the model predictions of Roussanov (2010) who shows that the relative status concerns associated with wealth are more important for the rich households, and thus, that the (status-seeking) rich optimally choose to invest in assets with highly idiosyncratic and positively skewed payoffs in order to advance their relative wealth position and get ahead of their peers.

Building on these findings, we next turn to the role of social status concerns of rich households and the induced lottery preferences in explaining our results. Specifically, Becker et al. (2000) argue that lottery-type preferences of status-seeking rich households are likely to be reflected in their investments through stocks, occupational choices, or entrepreneurial activity rather than simple lotteries, as the rich have the ability to gamble through utilizing such productive risks in the economy. To address this explanation, we adopt a revealed preference strategy and infer the lottery-type preferences of individuals from their investments in a particular type of financial security, that is, the Swedish Lottery Bonds. The lottery bonds are Swedish government bonds, which make coupon payments to bondholders based on lottery-like drawings and offer a (highly) positively skewed return distribution for their in-

¹ These explanations include liquidity (Bali and Cakici, 2008; Han and Lesmond, 2011), lottery demand (Bali et al., 2011), short-term reversal (Fu, 2009; Huang et al., 2010), average variance beta (Chen and Petkova, 2012), retail trading proportion (Han and Kumar, 2013), and arbitrage asymmetry (Stambaugh et al., 2015).

² Even conditional on participation, the direct stock investments of individuals are highly limited. For example, the average share of direct stock investments in the financial wealth of U.S. households represents only 5%.

vestors (Green and Rydqvist, 1997; 1999). Consistent with Becker et al. (2000), we find that moving up to the higher parts of the wealth distribution significantly increases the probability of households to invest in lottery bonds, suggesting that richer households tend to display stronger preferences for securities with lottery-type payoffs. More importantly, we find a significantly more pronounced positive effect of increases in household wealth on stock portfolio IVOL for households who are in the top of the wealth distribution and invest in lottery bonds. By contrast, we find no evidence of a significant effect in the sample of households who are in the bottom 90 percent of the wealth distribution. The documented asymmetry for the impact of lottery-type preferences on portfolio IVOL across wealthy and less-wealthy households suggests that social status concerns combined with the lottery-type preferences of the rich provide an explanation for their tendency to invest in high-volatility stocks. We also consider various alternative explanations, including family-controlled firms and insider holdings, constraints on financial leverage, and co-skewness risk, but find no empirical support for them.

We then investigate the implications of these household level findings on aggregate stock market outcomes. We first show that stocks with high IVOL indeed generate low risk-adjusted return (alpha) in the Swedish equity market where we base our empirical analysis, and that this negative cross-sectional relation between IVOL and future returns is largely driven by underperformance of the high-IVOL stocks. Furthermore, consistent with prior literature, we document that the negative alpha spread on IVOL-sorted portfolios is only significant for stocks predominantly held by retail investors. This suggests that the IVOL puzzle is primarily driven by individual investors' demand for high volatility stocks, which leads to temporary overpricing and negative future abnormal returns for such stocks.

Digging deeper, the bivariate portfolio-level analysis shows that the anomalous relation between IVOL and future returns is only confined to those stocks that are predominantly held by rich individuals, which mirrors our findings at the household level for the aggregate stock market outcomes. For example, the zero-cost portfolio that takes a long position in the high-IVOL stocks and a short position in the low-IVOL stocks that are largely held by rich investors generates a significantly negative and economically large alpha spread of -2.48% per month. In contrast, we find no evidence of the low risk anomaly for stocks with lower levels of rich investor ownership. These results are robust to the exclusion of large-cap, micro-cap, or popular stocks. We further show that the demand of rich households is distinct from that of non-rich households. For example, we find no significant cross-sectional relation between IVOL and future returns when considering the investments of the households in the middle or lower parts of the wealth distribution.

Then, we introduce a status-driven lottery demand factor and test if this newly proposed factor produces significant return and alpha spreads. We find that the new factor, constructed based on lottery demand and stock ownership

levels of rich individual investors, produces economically large and statistically significant premium after controlling for the market, size, value, and momentum factors. More importantly, this newly proposed factor explains the idiosyncratic volatility puzzle in Sweden. Thus, we conclude that social status concerns of rich households and the induced lottery preferences explain wealthy investors' demand for high-risk stocks, leading to overpricing and low future returns for such stocks.

Even though our portfolio analysis is standard in the literature, we recognize that a causal interpretation of the results from the bivariate portfolio-level analysis is subject to some limitations due to identification issues. To provide a more direct, causal interpretation of our results, we rely on a difference-in-difference type design where we focus on a subset of stocks, and exploit exogenous variation in the aggregate demand of rich households for individual stocks generated by a rule exception in inheritance and gift taxation of household stock wealth. During the 1999–2004 period, there was a large asymmetry in the inheritance and gift taxation of wealth held in stocks listed on different market segments. Specifically, large-cap stocks (i.e., A-listings) were valued at 75% of the stock price, while mid-/small-cap stocks (i.e., O-listings) were valued at 30% of their price at the time of assessment, which led to large differences in the inheritance and gift taxation of these stocks. Given this background, if a stock moves from the A-list to the O-list, this would create tax incentives among richer households, who are more prone to changes in inheritance and gift taxation, to rebalance their investments towards the switcher stocks. Exploiting the reasonably exogenous variation in rich households' aggregate demand for these stocks, we indeed find that an increase in rich households' demand for a particular stock leads to both an increase in the IVOL and significantly more negative risk-adjusted returns of that stock in the following month, even after controlling for stock and time fixed effects along with other time-varying stock characteristics. In summary, this analysis provides further support for the causal interpretation that it is the demand of rich households that partly drives the IVOL puzzle.

Our finding that the rich tend to hold stocks with high volatility is consistent with the results of previous studies (Calvet et al., 2007; Fagereng et al., 2020; Bach et al., 2020). For example, Calvet et al. (2007) show that households with higher financial wealth have increased their exposure to idiosyncratic risks. Using the same household panel as in our paper, Bach et al. (2020) find that the idiosyncratic volatility of returns to wealth increases in the net wealth of households, and is the highest for households in the top of the wealth distribution. Bach et al. (2020) also document that, despite having higher expected returns, rich households, including the top 0.01%, do not earn higher risk-adjusted returns on their financial wealth than the median household. Hence, the authors note that the rich do not possess superior investment skills than households in the middle or lower parts of the wealth distribution. Our results complement these studies by highlighting the cross-sectional asset pricing implications of the investment behavior of the rich, particularly, for the existence of the IVOL puzzle. Furthermore, we in-

investigate the economic underpinnings of the demand of wealthy households for high-volatility stocks.³

The remainder of the paper is organized as follows. [Section 2](#) describes the data, the sample construction, and provides summary statistics for the final sample. [Section 3](#) presents the empirical results of our household level analysis, where we study the stock investment behavior of rich households and discuss the potential motives for their tendency to invest in high-risk stocks. [Section 4](#) investigates the implications of our household level findings for aggregate stock market outcomes, particularly, for the existence of the IVOL puzzle. [Section 5](#) investigates whether the puzzling, negative relation between idiosyncratic volatility and future equity returns is significant in the cross-section of stocks trading in the U.S. and other international equity markets. [Section 6](#) concludes.

2. Data, sample construction, and descriptive statistics

In this section, we first describe the datasets and outline the key variables used in our empirical analyses. We then present descriptive statistics and provide a discussion on the stock investment behavior of the rich households.

2.1. Data and sample construction

When investigating the potential contribution of wealthy individuals to the IVOL puzzle, we focus on the Swedish stock market and utilize the Swedish Income and Wealth Registry, which is compiled by Statistics Sweden for tax purposes. The data provide highly accurate and detailed information on household debt and assets of every individual residing in Sweden between 1999 and 2007. It is important to note that it is the banks and financial institutions that report asset holdings to the tax authorities rather than individuals, which ensures the accuracy of the wealth information. The dataset provides information on the worldwide assets, disaggregated at the individual security or property level, held by residents as of December 31 of each year. Put differently, we are able to directly observe the detailed portfolio holdings of every resident, including the very rich, in Sweden at the individual stock or mutual fund level (e.g., how many shares of a given publicly-traded company a household holds in her individual stock portfolio). In addition to the wealth data, there is also detailed information about the demographics and labor market characteristics of all residents such as gender, age, level and type of education, labor income, place of residence, and industry of employment.⁴

In our analysis, we use information at the household rather than at the individual level. To identify the refer-

ence person (i.e., head of household) in a given household, we follow the Canberra definition.⁵ We use the socioeconomic characteristics of the household head when defining household controls such as age, gender, and industry of employment. The asset and debt holdings along with the income are aggregated at the household level.

When constructing the final sample, we proceed as follows. We restrict the sample to households who are aged between 18 and 70 in a given year between 1999 and 2007. We further eliminate those households who have either missing information on their disposable income or have an annual household disposable income less than 1,000 Swedish Kronor (SEK) (that is ≈ 100 US Dollars). Finally, we exclude from the sample households who have either zero financial or total wealth in any given year over the observation period.

In addition to the household panel data, we utilize auxiliary information on daily and monthly returns, shares outstanding, volume of shares, and balance sheet data of all companies listed on the Swedish equity market for the time period from January 1988 to December 2018 from the Thomson Reuters Datastream.⁶ Using this information, we calculate for each household in each year the stock portfolio returns and other portfolio characteristics such as the stock portfolio beta, idiosyncratic volatility (IVOL), size, book-to-market, and momentum over the sample period.

We then match the household data with the stock portfolio characteristics of the households. Note that the portfolio characteristics, such as IVOL or beta, are only constructed using the direct stock holdings of the households in Swedish companies. Hence, we implicitly impose another restriction in the sample construction, that is, we only focus on households who have positive holdings in Swedish single stocks in a given year.

Overall, our final sample includes approximately 10.5 million household-year observations, and around 1.8 million unique households for whom we have portfolio information. Note that we in addition exclude households who are in the top 0.1% of the wealth distribution from the sample to ensure that our results are not driven by the ultra-rich investors.⁷

⁵ See [Haliassos et al. \(2017\)](#) for further details.

⁶ Daily returns for each stock are calculated using a daily total return index which is adjusted for stock splits and dividend payments. We utilize returns denominated in US dollars. We also follow other international equity market studies such as [Bekaert et al. \(2007\)](#) and [Karolyi et al. \(2012\)](#) to screen the data and omit some of the data errors in Datastream that have been reported in the prior literature. Specifically, we select stocks only from the major exchange defined as that in which the majority of equities in Sweden are traded, and exclude stocks with special features such as depository receipts, real estate investment trusts, and preferred stocks. We retain all data for defunct stocks in the sample to avoid survivorship bias. We set the highest and lowest 1% of daily returns in each month to be missing and winsorize market value of equity at the 1% level in each month to deal with extreme observations. Moreover, monthly returns are winsorized at the 1% (99%) level for the left (right) tail for each month. To ensure that our results are not driven by penny stocks, we exclude stocks trading below \$1 per share.

⁷ For example, the mean net wealth of the households in the top 0.1% accounts for 75 million SEK with a standard deviation of 701 million SEK, suggesting that there is also substantial heterogeneity in the wealth of the ultra rich people.

³ Our paper also relates to the budding literature on the demand system approach in equity prices ([Koijen and Yogo, 2019](#); [Koijen et al., 2020](#)). For related applications on the role of individual investors and mutual funds in the cross-section of stock prices and asset pricing puzzles, see, for example, [Balasubramaniam et al. \(2023\)](#), [Betermier et al. \(2021\)](#) and [Han et al. \(2021\)](#), respectively.

⁴ The Swedish Income and Wealth Registry has been fruitfully used in earlier research for different purposes. See, e.g., [Calvet et al. \(2007\)](#), [Calvet et al. \(2009a\)](#), [Betermier et al. \(2017\)](#), and [Bach et al. \(2020\)](#) for a detailed description of the dataset.

Table 1

Household Demographic and Financial Characteristics.

This table reports the mean and standard deviation of the demographic and financial characteristics of the sampled households in our analysis for the period between 1999 and 2007. The unit of observation is household-year. The statistics for the full sample are based on 10,389,133 observations. The statistics for households in the top 10% and bottom 90% samples are based on 2,587,254 and 7,801,879 household-year observations, respectively. The values for income and wealth variables are in Swedish Krona (SEK) values (in 1000s). All monetary values are in 2007 values. See Data Appendix for detailed variable definitions.

	Full Sample		Top 10%		Bottom 90%	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
	(1)	(2)	(3)	(4)	(5)	(6)
Age	48.311	12.989	54.563	10.132	46.237	13.168
Male	0.677	0.467	0.706	0.456	0.668	0.471
College	0.421	0.494	0.486	0.499	0.399	0.489
Working for Government	0.214	0.41	0.201	0.401	0.219	0.413
Self-employed	0.075	0.264	0.124	0.329	0.059	0.237
Labor income (in 1000s)	395.204	309.331	477.811	381.658	367.810	275.861
Stock wealth (in 1000s)	185.602	814.869	539.069	1325.086	68.385	496.754
Financial wealth (in 1000s)	621.489	1280.162	1597.730	1952.888	297.749	704.769
Total Wealth (in 1000s)	2131.505	2658.449	5054.690	3560.804	1162.121	1197.047
Net Wealth (in 1000s)	1546.169	2057.166	4232.272	2432.443	655.405	697.898

2.2. Descriptive statistics

In this section we first provide some discussion on the demographic and financial characteristics of the sampled households, and further contrast the stock portfolio characteristics of the rich households with those of the non-rich households.

First, it is important to describe how we identify the rich households in the population. To do so, we first calculate the net wealth of each household by subtracting the household debt (i.e., mortgage loans and consumer debt) from total gross wealth that is the sum of all financial (i.e., direct and indirect stocks, bonds, and bank account holdings) and real assets (i.e., value of primary residence and other real estate holdings). We then compute the percentile rank of a given household in the entire wealth distribution in a given year.⁸ We define households who are in the top 10 percent of the net wealth distribution in each year as rich households.

Table 1 presents summary statistics for a select set of demographic and financial characteristics for our sample of households. In the first two columns of the table, we focus on the full sample. The middle and last two columns present descriptive statistics for households in the top 10 and bottom 90 percent of the wealth distribution, respectively. We observe that 68 percent of the sampled households is male-headed and the share of households with some college education accounts for 42 percent. 21 percent of the sample works in the public sector, and the average age is 48 years. When we contrast the demographic characteristics of the rich and non-rich households, we note that households in the top 10 percent of the wealth distribution tend to be older, better-educated and have significantly higher labor income than those in the bottom 90 percent of the wealth distribution.

In addition, the table reveals that the distribution of wealth across the population is highly skewed even though Sweden is characterized by relatively lower income inequality. For example, the top 10 share of net wealth accounts for 61.4% during the sample period, and the average net wealth of the rich households (mean value of approximately SEK 4.2 million) is 6.5 times larger than that of the households in the bottom 90 percent (mean value of SEK 0.65 million). A similar perspective applies to the stock wealth. Fig. 1 presents the share of direct stock wealth held by households who are in the top 10 percent of the wealth distribution over the sample period. We see that the top 10 share of direct stock wealth ranges between 80% and 88% with a time-series average of 83% during the 1999–2007 period, suggesting that stock wealth is highly concentrated in the top of the wealth distribution.

3. Do the rich gamble in the stock market?

One of the most puzzling empirical findings in the literature on portfolio choice is that only a small fraction of households (directly) invest in the stock market despite the equity premium (Mankiw and Zeldes, 1991; Haliassos and Bertaut, 1995). For example, in Sweden where the overall stock market participation rate is the highest in the world, less than one-third of households invest in individual stocks, and the average share of direct stock holdings in the gross wealth of the households represents only 3.9%. In an international context, these numbers are comparable, if anything, even lower.⁹ Put differently, the median house-

⁸ Note that we consider the entire population (i.e., not only the direct stockholders) when identifying the relative position of a household in the wealth distribution.

⁹ For example, the direct stock ownership rate in the U.S. is at around 14%. At the intensive margin, individual stocks represent, conditional on participation, only 5% of the financial wealth of the average U.S. household. The direct stock ownership rate in the Euro Area accounts for 8.2% with substantial heterogeneity across countries. For example, approximately 11% of German households are direct stockholders, whereas this number is only 3.1% in Italy. For further details, see Table O.A.1.

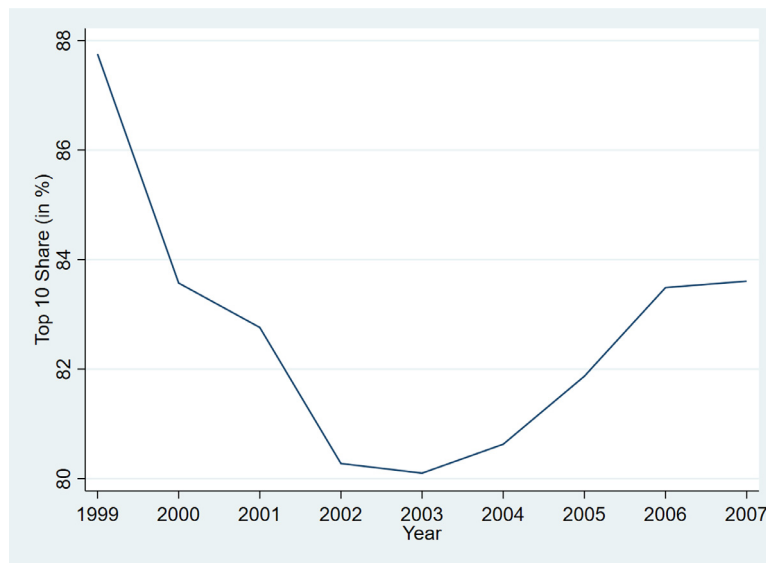


Fig. 1. Stock Wealth Inequality.

This figure depicts the share of direct stock wealth held by households who are in the top 10 percent of the wealth distribution in Sweden. The sample period is from 1999 to 2007. The y-axis reports the share of aggregate direct stock wealth (in %) held by households who are in the top 10 percent of the wealth distribution in a given year.

hold in most developed countries does not (directly) invest in the stock market.

Given this background, an essential question is how can so few individual investors affect equilibrium asset prices and contribute to the asset pricing puzzles, as proposed by the extensive literature in asset pricing? We argue that the highly skewed distribution of wealth across the population offers a potential explanation (Saez and Zucman, 2016; Piketty, 2015). In particular, if most of the wealth is concentrated in the hands of a small number of rich households, even a small subset of the population can affect equilibrium asset prices despite the low overall stock ownership rates. Consistent with this hypothesis, we indeed observe that household stock wealth, as compared to net wealth, is even more unequally distributed across the population. For example, in Sweden, households in the top 10 percent of the wealth distribution hold approximately 83% of the aggregate direct stock wealth, while this figure accounts for more than 87% in the U.S. To put these numbers into context, the top 10 share of aggregate net personal wealth in Sweden and in the U.S. accounts for 61% and 79% over the same period, respectively. Hence, it makes sense to study the portfolio selection behavior of the rich households (and their preferences) if one intends to understand the impact of individual investors on equilibrium stock prices, particularly, to the asset pricing anomalies.

To address this question, in what follows, we first use high quality household panel data to study the stock investment behavior of rich households with a particular focus on whether richer households tend to tilt their portfolios towards stocks with higher idiosyncratic volatility. We then discuss the potential motives of the wealthy households for their tendency to invest in high-risk stocks.

3.1. Baseline analysis

We begin our analysis by examining the potential relation between household wealth and exposure to idiosyncratic risk. In important contributions, Calvet et al. (2007) and Bach et al. (2020) focus on the full portfolio of risky assets and document that wealthier households tend to be more exposed to idiosyncratic risk, mainly because they invest more in individual stocks than in mutual funds. Regarding the latter, Bach et al. (2020) argue that the rich do so mainly to avoid the management fees charged by asset management companies and to develop investment strategies that better suit their needs and preferences. Despite this background, to the best of our knowledge, it is still largely unknown whether the (direct) stock portfolios of the rich have a higher or lower level of idiosyncratic risk than the stock portfolios of the less wealthy households.

To investigate this question, following Ang et al. (2006), we first calculate the idiosyncratic volatility (IVOL) of an individual stock (that is listed on the Swedish stock exchange) as the standard deviation of the residuals from time-series regressions of the daily excess stock returns on the daily excess market returns and the daily size (SMB) and book-to-market (HML) factor returns of Fama and French (1993) in a month. Since we only observe stock investments of sampled households on an annual frequency, we use the average IVOL of a stock in a given year.

We then compute, for each household in each year, the value-weighted direct stock portfolio IVOL as follows:

$$IVOL_{i,t}^P = \sum_{j=1}^N \overline{IVOL}_{j,t} \times \omega_{i,j,t} \quad (1)$$

where $IVOL_{i,t}^P$ is the stock portfolio IVOL of household i in year t , $\overline{IVOL}_{j,t}$ is the average IVOL of stock j in year t , and $\omega_{i,j,t}$ is the weight of stock j in the stock portfolio of household i in year t .

To formally analyze the relation between household wealth and household exposure to idiosyncratic risk, we regress the portfolio IVOL on net wealth levels or indicator variables for the rank of the household in the wealth distribution, controlling for a rich set of household and portfolio characteristics along with a battery of fixed effects. Specifically, the base regression model takes the following form:

$$IVOL_{i,t}^P = \beta_1 \cdot I\{Wealth_Rank\}_{i,t} + \beta_2 \cdot Controls_{i,t} + \gamma_i + \gamma_k + \gamma_l + \gamma_t + \epsilon_{i,t} \quad (2)$$

where $I\{Wealth_Rank\}_{i,t}$ is an indicator variable for the percentile rank of household i in the wealth distribution in a given year over the 1999–2007 period. In complementary analysis, we instead use the inverse hyperbolic sine function (IHS) of household net wealth as the key variable of interest. This transformation is similar to a log-transformation, and hence, the estimates can be interpreted as semi-elasticities. $Controls_{i,t}$ is a vector of household and portfolio characteristics, which includes the (IHS of) household income, age of the household head, an indicator variable for having college education, and portfolio level controls such as the average size, book-to-market, illiquidity, and momentum of the stock portfolio. In the base regressions, we also control for household fixed effects, γ_i , regional fixed effects, defined at the municipality level and denoted by γ_k , to account for potential differences in regional economic conditions,¹⁰ and fixed effects for industry of employment, γ_l , and year of observation, γ_t , respectively. The standard errors are clustered at the household level.

Since our regressions control for household fixed effects, we are able to account for all sources of unobserved, time-invariant household characteristics. Hence, our identification comes entirely from the within-household variation. In other words, we determine the impact of becoming richer or moving into the top of the wealth distribution on changes in household exposure to IVOL.

Table 2 presents the regression results where we first use the IHS of household net wealth as key variable of interest. In column (1), we focus on the entire universe of stockholding households. Columns (2) and (3) report the regression results separately for households who are in the bottom 90 and top 10 percent of the wealth distribution, respectively.

The table reveals an interesting pattern. The regression estimates in column (1) show a negative and significant effect of an increase in wealth on portfolio IVOL of households. Put differently, when households become wealthier, they tend to lower their idiosyncratic risk exposure as measured by the IVOL of their stock portfolio, on average. In fact, this finding is consistent with the results of

Table 2

The Effect of Household Wealth on Portfolio IVOL.

This table presents coefficient estimates from panel regressions for stock portfolio IVOL. In all specifications, portfolio IVOL is regressed on (IHS of) household net wealth, observable household variables, and a battery of fixed effects. All regressions account for time-year, household, municipality of residence, and industry of employment fixed effects. In column (1), we focus on the entire universe of stockholding households. Columns (2) and (3) report the regression results separately for households who are in the bottom 90 and top 10 percent of the wealth distribution, respectively. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL (%)		
	Full (1)	Bottom 90% (2)	Top 10% (3)
(IHS of) Net Wealth	−0.0011 (−27.89)	−0.0010 (−25.59)	0.0392 (17.32)
Income	−0.0102 (−31.26)	−0.0122 (−29.58)	−0.0047 (−8.58)
College	−0.0176 (−12.25)	−0.0203 (−12.04)	−0.0049 (−1.74)
Age	0.0005 (2.65)	0.0002 (1.05)	0.0014 (3.91)
Portfolio: Size	−0.1576 (−521.72)	−0.1601 (−458.15)	−0.1534 (−240.37)
Portfolio: Liquidity	65.8739 (384.12)	65.2540 (328.55)	65.7135 (183.16)
Portfolio: BM	−0.7603 (−451.87)	−0.7289 (−372.83)	−0.8323 (−235.36)
Portfolio: Momentum	−0.1460 (−237.51)	−0.1553 (−218.81)	−0.0989 (−72.94)
Constant	4.0769 (387.93)	4.1368 (332.21)	3.2598 (79.53)
Obs	8,710,090	6,572,858	2,027,091
FE: time	Yes	Yes	Yes
FE: municipality	Yes	Yes	Yes
FE: industry	Yes	Yes	Yes
FE: household	Yes	Yes	Yes

Kumar (2009) who documents a negative correlation between household wealth and weight of lottery-type stocks in the equity portfolio of retail investors. When we restrict our sample to households who are in the bottom 90 percent of the wealth distribution, we again find a similar pattern; i.e., there is a significantly negative relation between household wealth and stock portfolio IVOL. Interestingly, when we re-run the same analysis but only including households who are in the top of the wealth distribution, the impact of wealth on portfolio IVOL reverses, as presented in column (3). Specifically, we now observe that changes in household wealth display a positive and statistically highly significant effect on the tendency of households who rank high in the wealth distribution to take on more idiosyncratic risk. This finding suggests that wealth tends to exhibit opposite effects on the idiosyncratic risk taking behavior of households who are in different social classes, as measured by their relative position in the wealth distribution.

Controlling for other household characteristics, we find that household income and having some college education both display a negative effect on stock portfolio IVOL of households in the full sample as well as in the sample splits by wealth rank. The former finding is interesting, and suggests that the reversal in the effect that we doc-

¹⁰ In Sweden, there are 290 municipalities that are responsible for various tasks such as social services or physical planning. Hence, municipality fixed effects account for latent regional characteristics and capture the direct effects of the location of households.

Table 3

The Effect of Household Wealth on Portfolio IVOL: Non-linear Effects. This table presents coefficient estimates from panel regressions for stock portfolio IVOL. In all specifications, portfolio IVOL is regressed on indicator variables for household rank in the wealth distribution, observable household variables, and a battery of fixed effects. All regressions account for time-year, household, municipality of residence, and industry of employment fixed effects. In these regressions, we focus on the entire universe of stockholding households. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL (%)	
	(1)	(2)
$I\{P90 - P100\}$	0.0066 (6.69)	
$I\{P90 - P95\}$		0.0039 (3.87)
$I\{P95 - P99\}$		0.0155 (10.82)
$I\{P99 - P100\}$		0.0421 (16.39)
Income	-0.0105 (-32.12)	-0.0105 (-32.38)
College	-0.0161 (-11.19)	-0.0160 (-11.13)
Age	0.0005 (2.81)	0.0006 (2.88)
Portfolio: Size	-0.1576 (-521.84)	-0.1576 (-521.88)
Portfolio: Liquidity	65.8848 (384.19)	65.8896 (384.23)
Portfolio: BM	-0.7600 (-451.76)	-0.7597 (-451.56)
Portfolio: Momentum	-0.1463 (-237.96)	-0.1464 (-238.15)
Constant	4.0654 (387.04)	4.0642 (386.97)
Obs	8,710,090	8,710,090
FE: time	Yes	Yes
FE: municipality	Yes	Yes
FE: industry	Yes	Yes
FE: household	Yes	Yes

ument across wealthy and relatively poorer households is only confined to changes in wealth but not to household income.

To refine our understanding on the potential non-linear effects of wealth on household risk taking, we next focus on changes in the relative position of households in the wealth distribution, and re-estimate the portfolio regressions. In column (1) of Table 3, we first define an indicator variable that takes the value of 1 if the household is in the top 10 percent of the wealth distribution in a given year. In column (2), we use separate variables for different ranks of households in the top of the wealth distribution such as indicator variables for being in the 95th to 99th wealth percentiles or being in the top 1 percent of the wealth distribution. Note that we consider the entire universe of stockholding households in these regressions, and again control for all the relevant household and portfolio characteristics and fixed effects, as shown in Eq. (2).

The regression estimates in Table 3 confirm the patterns observed in the sample splits. Specifically, we observe that households who move up to the top 10 percent of the wealth distribution significantly increase their stock port-

folio exposure to IVOL. As shown in column (2), we also find that the documented effect increases with the relative wealth rank of households in a non-linear fashion. For example, moving up to the top 1 percent of the wealth distribution displays four times larger effect on portfolio IVOL than moving into the 95th to 99th wealth percentiles. Hence, these results highlight the significant non-linear effects of wealth on household portfolio choice, particularly, on their exposure to idiosyncratic risk. In this sense, our results are in line with the social status model of Roussanov (2010) that predicts that risky investments with highly idiosyncratic payoffs can be particularly attractive to status-seeking rich households if the satisfaction brought by "getting ahead of the Joneses" dominates the risk of falling behind. Specifically, Roussanov (2010) shows that the relative status concerns associated with wealth are more important for the rich households, and their marginal utility of wealth increases when they attain higher relative wealth rank (Friedman and Savage, 1948). Hence, the rich are more inclined to undertake greater private risk through investing in stocks with higher idiosyncratic risk, in an objective not just to keep up or catch up with their peers, but actually to get ahead of them. We discuss this particular mechanism in detail in the next section.

One could argue that the documented positive relationship between household wealth and exposure to IVOL in the stock portfolio of rich households may partly be driven by passive changes in portfolio weights induced by return realizations. We address this concern in several ways. First, we repeat the same analysis as in Tables 2 and 3, but exclude total equity from the definition of net wealth. In other words, we focus only on changes in household net wealth, net of any changes in the value of the stock portfolio. As reported in Tables O.A.2 and O.A.3 of the online appendix, we reassuringly find similar results. Second, we identify households' new stock positions, i.e., stocks that households did not own at the end of the previous year but purchased at some point during the current year. We then compute the value-weighted portfolio IVOL but focus only on newly purchased stocks and repeat the analysis. The results reported in Tables O.A.4 and O.A.5 using this alternative sample confirm our baseline results in Tables 2 and 3. Finally, following Calvet et al. (2009b), we first measure the degree of risky share inertia, that is measured as the absolute value of the change in the risky portfolio share, for each household and then use this value to split the households in the sample into two groups: (i) active rebalancers, and (ii) inert households. We define active rebalancers (inert households) as those with a below (above) median degree of portfolio inertia over the sample period. We then focus on the sample of active rebalancers and re-run the analysis. The regression results, tabulated in Tables O.A.6 and O.A.7, show that households moving to the top of the wealth distribution significantly increase their equity portfolio exposure to IVOL, even among the sample of households that rebalance their portfolios more actively. Overall, these additional sensitivity analyses suggest that households actively increase their exposure to idiosyncratic risk as they become wealthier.

Up until now in our analysis, we account for various sources of time-invariant omitted characteristics at

Table 4

The Effect of Household Wealth on Portfolio IVOL: High-dimensional Fixed Effects.

This table presents coefficient estimates from panel regressions for stock portfolio IVOL. In all specifications, portfolio IVOL is regressed on indicator variables for household rank in the wealth distribution, observable household variables, and a battery of fixed effects. All regressions account for household fixed effects. Columns (1) and (2) introduce municipality-by-year and industry-by-year fixed effects, respectively. In column (3), we control for both municipality-by-year and industry-by-year fixed effects. In these regressions, we focus on the entire universe of stockholding households. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL (%)		
	(1)	(2)	(3)
$I\{P90 - P95\}$	0.0029 (3.18)	0.0068 (6.85)	0.0070 (7.08)
$I\{P90 - P95\}$	0.0138 (10.67)	0.0181 (12.82)	0.0184 (13.02)
$I\{P99 - P100\}$	0.0407 (17.60)	0.0373 (14.67)	0.0372 (14.64)
Income	-0.0026 (-31.31)	-0.0078 (-24.28)	-0.0067 (-20.86)
College	-0.0149 (-11.61)	-0.0146 (-10.21)	-0.0128 (-9.00)
Age	0.0002 (1.34)	0.0003 (1.58)	0.0001 (0.47)
Portfolio: Size	-0.1558 (-549.19)	-0.1580 (-524.47)	-0.1585 (-526.80)
Portfolio: Liquidity	66.3822 (409.75)	65.3388 (379.46)	65.0806 (376.73)
Portfolio: BM	-0.7479 (-482.20)	-0.7589 (-448.53)	-0.7603 (-447.35)
Portfolio: Momentum	-0.1438 (-252.45)	-0.1411 (-224.89)	-0.1382 (-218.88)
Constant	3.9333 (461.73)	4.0411 (389.25)	4.0400 (392.88)
Obs	10,199,966	8,710,089	8,710,089
FE: time	No	No	No
FE: household	Yes	Yes	Yes
FE: municipality-year	Yes	No	Yes
FE: industry-year	No	Yes	Yes

the household, region, and industry of employment levels, which could otherwise generate a spurious correlation between household wealth and the tendency of households to invest in high-IVOL stocks. Still, it is possible that there are some time-varying unobserved shocks either at the region- or industry-level that could make households richer, and at the same time, make them more likely to invest in high-IVOL stocks. This would compromise our identification strategy. To account for such unobserved shocks, we next augment the regression specification in Eq. (2) with municipality-year and industry-year fixed effects. As reported in Table 4, we obtain similar, if anything, stronger results when we introduce these fixed effects in our baseline analysis. These findings further alleviate any concerns that our results are driven by any time-varying regional or industry-level unobserved shocks.

In Tables O.A.8–O.A.13 of the online appendix, we perform several additional sensitivity checks such as using alternative sample constructions; considering the ultra-rich households who are in the top 0.1% of the wealth distribution; controlling for additional household observable characteristics, such as the content of education or occupation,

in the regressions; using alternative regression specifications; excluding self-employed households; and conducting additional tests to ensure that the statistical inference in our base analysis is not sensitive to potential downward biases in standard errors. All these tests confirm our conclusions from the base analysis and further ensure the robustness of our key findings.

Taken together, the analysis presented in this section indicates that households who become wealthier and move up to higher social classes, as measured by their relative position in the wealth distribution, significantly increase their stock portfolio exposure to idiosyncratic risk.

3.2. Why do the rich gamble in the stock market?

In this section, we investigate potential economic mechanisms underlying the relationship between the demand of wealthy investors and the low risk anomalies.

3.2.1. Status and lottery-type preferences of rich households

One potential explanation for the documented tendency of wealthier investors to invest relatively more in stocks with higher idiosyncratic volatility is their non-standard preferences. For example, Roussanov (2010) shows that rich households who are more concerned about their social status can optimally choose to invest in assets with highly idiosyncratic and positively skewed payoffs in order to advance their relative wealth position and get ahead of their peers. Our findings in the previous section are in fact consistent with these predictions of the status model of Roussanov (2010), that is, the positive effect of wealth on portfolio IVOL is only operative for those households who become wealthier and move up to the top of the wealth distribution where relative status concerns are more important. At the same time, several papers suggest that status-seeking individuals tend to exhibit a stronger propensity for lottery-like features (Brunk, 1981; Becker et al., 2000; Clotfelter and Cook, 1991). In particular, Becker et al. (2000) argue that lottery-type preferences of status-seeking rich households are likely to be reflected in their investments through stocks, occupational choices, or entrepreneurial activity rather than simple lotteries, as the rich have ability to gamble through utilizing such productive risks in the economy. Given this background, we next analyze the role of social status concerns of rich households and the induced lottery preferences in explaining our findings.

First, we acknowledge that it is a daunting task to directly elicit the preferences of individuals using non-experimental data. Even though our household level dataset is very rich in many dimensions, it does not provide any information on elicited preferences or beliefs of households. Therefore, we adopt a revealed preference strategy and infer the lottery-type preferences of individuals from their financial behavior. Specifically, we consider the investments of households in a particular type of financial security, that is, the Swedish Lottery Bonds. The lottery bonds are Swedish government bonds, which make coupon payments to bondholders on the basis of lottery-like drawings (Green and Rydqvist, 1997). In other words, coupons received by the bondholders are purely random

Table 5

The Effect of Household Wealth on Lottery Bond Participation.

This table presents coefficient estimates from panel regressions for lottery bond participation. In all specifications, the indicator variable for lottery bond participation is regressed on indicator variables for household rank in the wealth distribution, observable household variables, and a battery of fixed effects. In columns (1) and (2), we focus on the entire universe of stockholding households, while, in columns (3) and (4), we restrict the sample to those stockholding households who also invest in bonds directly (other than lottery bonds). In all regressions, we account for household, municipality, time-year, and industry fixed effects. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	Lottery Bond Participation			
	(1)	(2)	(3)	(4)
$I\{P90 - P100\}$	0.0163 (31.70)		0.0185 (17.17)	
$I\{P90 - P95\}$		0.0138 (26.20)		0.0155 (14.20)
$I\{P90 - P95\}$		0.0259 (33.90)		0.0317 (20.59)
$I\{P99 - P100\}$		0.0337 (25.38)		0.0431 (17.25)
Income	-0.0011 (-6.34)	-0.0011 (-6.69)	-0.0014 (-3.47)	-0.0016 (-3.88)
College	-0.0019 (-3.02)	-0.0018 (-2.92)	-0.0009 (-0.53)	-0.0008 (-0.49)
Age	-0.0008 (-8.73)	-0.0008 (-8.61)	-0.0004 (-1.60)	-0.0003 (-1.51)
Constant	0.1733 (35.83)	0.1724 (35.65)	0.2392 (19.32)	0.2373 (19.17)
Obs	8,710,090	8,710,090	2,068,618	2,068,618
FE: time	Yes	Yes	Yes	Yes
FE: household	Yes	Yes	Yes	Yes
FE: municipality	Yes	Yes	Yes	Yes
FE: industry	Yes	Yes	Yes	Yes

(Green and Rydqvist, 1999). Overall, these financial instruments offer a (highly) positively skewed return distribution for their investors.

Using this information, we first ask whether households who climb up the wealth distribution and have status concerns display a stronger tendency to "gamble" through investing in lottery bonds, as argued by Becker et al. (2000). To answer this question, we define an indicator variable that takes the value of 1 if a household holds lottery bonds in a given year, and 0 otherwise. We then regress the lottery bond participation variable on indicator variables for the relative wealth rank of households, other household and portfolio controls along with a battery of fixed effects.

The results are reported in Table 5. Consistent with Becker et al. (2000), we document that moving up to the upper parts of the wealth distribution significantly increases the probability of households to invest in lottery bonds, regardless of the definition of the relative wealth position of households. For example, based on the regression estimates in column (2), moving to the top 1 percent of the wealth distribution increases the likelihood to invest in lottery bonds by 3.4 percentage points (t-stat. = 25.38). The effect is not only statistically significant, but also highly meaningful in economic terms. In particular, it implies a 27% increase in lottery bond participation, as the mean participation rate in our sample equals 12.4 percent. Interestingly, we also observe that the magnitude of this effect monotonically increases in the relative wealth rank of households, suggesting that households who attain higher relative positions in the wealth distribution, and

possibly value status more tend to display stronger preferences for securities with lottery-type payoffs.¹¹

We recognize the possibility that the documented positive effect of wealth on lottery bond participation could be partly driven by the greater ability of the rich to pay the transaction costs of directly investing in bonds. To address this concern, we next focus on a subsample of households holding bonds directly (other than lottery bonds) and repeat the analysis. As presented in columns (3) and (4) of Table 5, we find similar results, suggesting that participation costs alone do not explain our findings.

To further scrutinize this mechanism, we next estimate an interaction model where we augment the portfolio-level IVOL regressions by incorporating an interaction term between household net wealth and share of lottery bonds in household financial wealth in our estimation model. Table 6 presents the regression estimates for a sample split of households based on their relative po-

¹¹ As an alternative way of measuring the preferences of wealthy households for securities with lottery-type preferences, we compute the portfolio MAX for each household in the sample. Specifically, following Bali et al. (2011), we measure the lottery payoff of each stock by the average of the five highest daily returns in each month. Since we only observe the stock investments of the sampled households on an annual frequency, we use the average MAX of a stock in a given year. We then compute the value-weighted direct stock portfolio MAX for each household in each year and regress this variable on household and portfolio characteristics along with a battery of fixed effects. As presented in Table O.A.14 of the online appendix, we find that households who move up to higher ranks in the wealth distribution significantly increase their exposure to stocks with lottery-like payoffs as measured by their MAX.

Table 6

The Effect of Household Wealth on Portfolio IVOL: The Role of Lottery-type Preferences.

This table presents coefficient estimates from panel regressions for stock portfolio IVOL. In all specifications, portfolio IVOL is regressed on (IHS of) household net wealth, the share of lottery bonds in the portfolio, an interaction of household net wealth and the share of lottery bonds in the portfolio, observable household variables, and a battery of fixed effects. All regressions account for time-year, household, municipality of residence, and industry of employment fixed effects. Columns (1) and (2) report the regression results separately for households who are in the bottom 90 and top 10 percent of the wealth distribution, respectively. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL (%)	
	Bottom 90% (1)	Top 10% (2)
(IHS of) Net Wealth	−0.0010 (−25.41)	0.0376 (16.52)
Share Lottery Bond	−0.0484 (−4.69)	−5.7070 (−6.08)
Share Lottery Bond x (IHS of) Net Wealth	−0.0010 (−1.39)	0.3493 (5.90)
Income	−0.0122 (−29.63)	−0.0047 (−8.50)
College	−0.0203 (−12.05)	−0.0049 (−1.75)
Age	0.0003 (1.06)	0.0014 (3.92)
Portfolio: Size	−0.1601 (−458.16)	−0.1534 (−240.39)
Portfolio: Liquidity	65.2535 (328.55)	65.7139 (183.19)
Portfolio: BM	−0.7289 (−372.83)	−0.8323 (−235.35)
Portfolio: Momentum	−0.1554 (−218.83)	−0.0989 (−72.95)
Constant	4.1375 (332.25)	3.2846 (79.71)
Obs	6,572,858	2,027,091
FE: time	Yes	Yes
FE: municipality	Yes	Yes
FE: industry	Yes	Yes
FE: household	Yes	Yes

sition in the wealth distribution. Specifically, columns (1) and (2) consider households who are in the bottom 90 percent and top 10 percent of the wealth distribution, respectively.

Similar to the model predictions of [Becker et al. \(2000\)](#) and [Roussanov \(2010\)](#), we document a significantly more pronounced positive effect of household wealth on stock portfolio IVOL for those households who are in the top of the wealth distribution and invest relatively more in lottery bonds. By contrast, we find no evidence of a significant effect in the sample of households who are in the bottom 90 percent of the wealth distribution. As presented in Table O.A.15 of the online appendix, we obtain similar results when we instead use an indicator variable for lottery bond participation.

Taken together, the documented asymmetry in the impact of lottery-like preferences on portfolio IVOL between wealthy and less wealthy households suggests that social status concerns of rich households and the induced preferences for lottery-like securities play an important role in their tendency to hold stock portfolios with higher levels of idiosyncratic risk.

3.2.2. Robustness to alternative explanations

In what follows, we investigate alternative sources of high demand by richer households for stocks with higher

idiosyncratic risk and test whether these potential driving forces can provide a partial or full explanation for our empirical results.

One could argue that our empirical findings, at least partially, may be driven by the insider holdings of wealthy "industrialists" in family-owned firms ([La Porta et al., 1999](#); [Faccio and Lang, 2002](#); [Franks et al., 2012](#)). We address this alternative interpretation in two ways. First, we make use of auxiliary data provided by [Faccio and Lang \(2002\)](#) for Swedish companies, which provide detailed information on the ownership structure (i.e., ultimate ownership and control) of all listed firms in Sweden. Following the standard definition in the literature (e.g., [La Porta et al., 1999](#) and [Faccio and Lang, 2002](#)), we use the 20% voting shares threshold when defining ownership control, and define those firms as family-controlled if the ultimate owner of a given company is an individual, a family or a firm that is unlisted on any stock exchange. According to this definition, as of 1998 (i.e., this is the year when the ownership structure data were collected), [Faccio and Lang \(2002\)](#) identify 56 out of 245 firms in Sweden as family-controlled. To alleviate any concerns that our results may be a statistical artifact of the holdings of the rich in their family-owned companies, we next repeat our analysis excluding these family firms from the sample of stocks. As tabulated in columns (1) and (2) of [Table 7](#), despite the

Table 7

The Effect of Household Wealth on Portfolio IVOL: Alternative Explanations.

This table presents coefficient estimates from panel regressions for stock portfolio IVOL. In all specifications, portfolio IVOL is regressed on (IHS of) household net wealth, observable household variables, and a battery of fixed effects. All regressions account for time-year, household, municipality of residence, and industry of employment fixed effects. Columns (1), (3), (5), and (7) report the regression results households who are in the bottom 90, and columns (2), (4), (6), and (8) report the results for households in the top 10 percent of the wealth distribution, respectively. Standard errors are clustered at the household level. The corresponding t-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL (%)							
	Family-firms		Insider Holdings		Leverage-constraints		Co-skewness	
	Bottom 90% (1)	Top 10% (2)	Bottom 90% (3)	Top 10% (4)	Bottom 90% (5)	Top 10% (6)	Bottom 90% (7)	Top 10% (8)
(IHS of) Net Wealth	−0.0010 (−26.19)	0.0372 (16.28)	−0.0010 (−25.96)	0.0369 (16.72)	−0.0010 (−20.41)	0.0342 (13.58)	−0.0010 (−24.26)	0.0433 (19.19)
Income	−0.0125 (−30.15)	−0.0046 (−8.29)	−0.0118 (−29.25)	−0.0044 (−8.15)	−0.0127 (−28.70)	−0.0051 (−8.57)	−0.0124 (−30.22)	−0.0050 (−9.05)
College	−0.0204 (−11.97)	−0.0047 (−1.66)	−0.0204 (−12.27)	−0.0043 (−1.57)	−0.0226 (−12.52)	−0.0037 (−1.22)	−0.0205 (−12.22)	−0.0048 (−1.72)
Age	0.0003 (1.26)	0.0014 (4.10)	0.0002 (1.07)	0.0013 (3.86)	−0.0001 (−0.36)	0.0010 (2.61)	0.0002 (0.99)	0.0014 (3.98)
Portfolio: Size	−0.1545 (−446.27)	−0.1463 (−232.86)	−0.1497 (−430.35)	−0.1378 (−216.24)	−0.1580 (−411.62)	−0.1519 (−220.02)	−0.1709 (−481.21)	−0.1651 (−254.36)
Portfolio: Liquidity	68.7775 (332.56)	69.4825 (186.06)	79.1823 (299.03)	79.8558 (157.30)	65.8090 (297.87)	65.9286 (167.44)	62.2876 (314.00)	62.9468 (176.63)
Portfolio: BM	−0.7412 (−376.78)	−0.8423 (−238.97)	−0.7202 (−375.91)	−0.8127 (−233.97)	−0.7171 (−346.36)	−0.8289 (−221.97)	−0.7110 (−361.88)	−0.8170 (−230.65)
Portfolio: Momentum	−0.1525 (−210.80)	−0.1022 (−75.22)	−0.1574 (−224.21)	−0.1051 (−78.72)	−0.1573 (−206.34)	−0.0896 (−61.77)	−0.2140 (−230.77)	−0.1535 (−90.27)
Portfolio: Coskewness							0.0119 (161.14)	0.0124 (90.53)
Constant	4.1059 (327.10)	3.2411 (78.42)	4.0278 (328.12)	3.1373 (78.31)	4.1277 (301.74)	3.3410 (73.53)	4.2204 (340.29)	3.2890 (80.55)
Obs	6,493,242	2,015,635	6,567,837	2,026,390	5,711,582	1,775,445	6,572,858	2,027,091
FE: time	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE: municipality	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE: industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE: household	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

slight decline in magnitudes as compared to our base estimates, we still observe both statistically and economically significant positive effects of wealth on portfolio IVOL only among households who are in the top 10 percent of the wealth distribution.

Second, following [Bach et al. \(2020\)](#), we exclude from the sample those individual household stock holdings that are larger than 5% of a company's market capitalization, and re-run the portfolio regressions. Columns (3) and (4) of [Table 7](#) show a significantly positive effect of wealth on portfolio IVOL only for the richer households, while we still observe a negative relation between wealth and portfolio IVOL for the less-wealthy households. Overall, these sensitivity checks indicate that our findings are not driven by the insider holdings of rich households in family-owned companies.

Third, we examine whether our findings may be explained by constraints on rich investors' ability to take on leverage.¹² To test this hypothesis, we first compute for each sampled household the leverage ratio, which is calculated as the ratio of household debt outstanding to annual

household disposable income in a given year. Then, we exclude (rich) households with a leverage ratio that exceeds the 90th percentile of the leverage distribution in a year, and re-run our analysis. As reported in columns (5)–(6) of [Table 7](#), we obtain similar results to what we observe in the baseline analysis. Thus, we conclude that the tendency of rich households to invest in stocks with high IVOL is not driven by constraints on financial leverage of rich households.

Fourth, we test whether household exposure to co-skewness risk may provide an explanation for our results.¹³ Specifically, we first estimate for the Swedish equities the physical measure of co-skewness following [Harvey and Siddique \(2000\)](#).¹⁴ Then, we calculate the portfolio co-skewness at the household level, and include it as an additional control in our regressions. The results are reported in the last two columns of [Table 7](#). We find that the

¹² [Garleanu and Pedersen \(2011\)](#) develop a model in which constraints on investors' ability to take on leverage (funding liquidity constraints) affect equilibrium asset prices. Also, [Frazzini and Pedersen \(2014\)](#) propose a model with leverage and margin constraints that vary across investors and time.

¹³ [Schneider et al. \(2020\)](#) show that the low risk anomalies arise when investors require compensation for co-skewness risk. They find that option-implied ex-ante skewness is strongly related to ex-post residual co-skewness, which is directly linked to a stock's CAPM beta. Thus, the CAPM alphas reflect compensation for residual co-skewness risk.

¹⁴ Because we do not have access to optionable stocks trading in the Swedish equity market, we cannot estimate the option-implied measure of co-skewness used by [Schneider et al. \(2020\)](#). Thus, we rely on the physical measure of co-skewness originally proposed by [Harvey and Siddique \(2000\)](#).

coefficient on household net wealth retains its statistical and economic significance even when we explicitly control for portfolio co-skewness, suggesting that systematic co-skewness risk does not fully explain our results.

Finally, to check whether our results can be generalized to other low risk anomalies, we focus on different stock characteristics (other than IVOL) that are documented to generate an anomalous risk-return relation. These include market beta (BETA), Value-at-Risk (VaR) and Expected Shortfall (ES) to proxy for downside risk. First, contrary to the fundamental principle that higher market risk (beta) is compensated with higher expected return, [Frazzini and Pedersen \(2014\)](#) show that high-beta stocks underperform low-beta stocks.¹⁵ Second, [Campbell et al. \(2008\)](#) provide evidence that financially distressed stocks with a high probability of default deliver anomalously low future returns. [Campbell et al. \(2008\)](#) also show that stocks with higher default risk offer returns with higher standard deviation, higher market beta, and higher positive skewness than stocks with lower default risk.¹⁶

We construct portfolio level measures for BETA, ES, and VaR for each sampled household, analogous to the way we measure portfolio IVOL at the household level as shown in [Eq. \(1\)](#). Then, we regress each of these measures on (IHS of) household net wealth or on indicator variables for the rank of the household in the wealth distribution, controlling for a rich set of household and portfolio characteristics along with a battery of fixed effects. The regression estimates are presented in Tables O.A.16 and O.A.17 of the online appendix. Similar to the portfolio IVOL results, we observe that households who attain higher rank in the wealth distribution significantly increase their exposure to high-risk stocks, as measured by their BETA, ES, or VaR, respectively. Hence, it seems that our empirical findings extend beyond the IVOL to other low risk anomalies.

4. Rich households and the IVOL puzzle

The household level analysis in the previous section shows that households who become richer and move up to higher social classes, as measured by their relative position in the wealth distribution, tend to significantly increase their stock portfolio exposure to idiosyncratic risk. In this section, we investigate the implications of these findings for aggregate stock market outcomes, particularly, for the existence of the IVOL puzzle.

¹⁵ We follow [Scholes and Williams \(1977\)](#) to estimate the market beta for each stock and each month in our sample to account for nonsynchronous trading.

¹⁶ We proxy downside risk with the 5% VaR measure of [Atilgan et al. \(2020\)](#). The 5% VaR is calculated as the 5th percentile of daily returns over the past year (250 trading days) at the end of each month, with the constraint that there should be at least 200 non-missing return observations over the past year. Since the maximum likely loss values obtained from the left tail of the return distribution are negative, we multiply the 5th percentile values by -1 , so that higher VaR values correspond to higher levels of downside risk. Finally, ES is defined as the average of the observations that are less than or equal to the 5th percentile of the daily returns for each stock during the past year as of the end of month t .

4.1. The IVOL puzzle in Sweden

We begin our analysis by investigating the existence and significance of the IVOL puzzle in the Swedish equity market. To do so, for each month from January 1988 to December 2018, we first sort stocks trading in the Swedish equity market into decile portfolios based on their IVOL, where decile 1 (decile 10) contains stocks with the lowest (highest) IVOL. Following earlier studies on idiosyncratic volatility (e.g., [Ang et al., 2006](#); [Kumar, 2009](#); [Bali et al., 2011](#)), we then calculate the one-month-ahead value-weighted average portfolio returns, and this procedure is repeated each month until the sample is exhausted.

[Table 8](#) reports for each decile the average IVOL, the next-month average excess return, and the risk-adjusted returns (alphas) based on the global [Asness and Frazzini \(2013\)](#) model with the global market, size, value, and momentum factors (i.e., AF4 Alpha) constructed using stocks trading in 22 developed countries. The last column of [Table 8](#) presents the 10-1 average IVOL, excess return, and risk-adjusted return spreads for the arbitrage portfolio that is long in the decile of stocks with the highest IVOL and short in the decile of stocks with the lowest IVOL.¹⁷

As presented in Panel A of [Table 8](#), univariate portfolio sorts of Swedish equities indicate a significantly negative relation between IVOL and next-month average returns; the arbitrage portfolio (High-Low IVOL) loses on average 2.90% per month with a Newey-West t -statistic of -4.28 .¹⁸ We obtain similar results based on the risk-adjusted returns from the global [Asness and Frazzini \(2013\)](#) model; the AF4 alpha spread for the long-short portfolio account for -2.41% per month with a t -statistic of -4.01 . Thus, we conclude that the IVOL puzzle is not explained by the global market, size, value, and momentum factors. In additional sensitivity checks, we reconstruct the IVOL-sorted portfolios by excluding illiquid, small, and large stocks. As shown in Panel B of [Table 8](#), the IVOL puzzle is not driven by large popular stocks, costly arbitrage, or liquidity risk.

We also investigate the source of the economically large risk-adjusted return spreads between the high- and low-IVOL portfolios: Is it due to outperformance by low-IVOL stocks, underperformance by high-IVOL stocks, or both? For this, we focus on the economic and statistical significance of the risk-adjusted returns of decile 1 versus decile 10 of the value-weighted portfolios. As reported in [Table 8](#), we observe that the overpricing of the high-IVOL stocks appears to be operative for the Swedish equities. The AF4 alpha of decile 10 (that are high-IVOL stocks) is negative and account for -2.05% (t -stat. = -3.09) per month, whereas the corresponding alpha of decile 1 (low-IVOL

¹⁷ The Swedish stock market is mainly dominated by a small number of very large companies ([Betermier et al., 2017](#)). To alleviate any concerns regarding the effects of large companies in our analysis, we repeat the univariate sorts by excluding those companies that are in the top 1 or top 5 or top 10 percent of the size distribution, and obtain similar results. In our preferred specification, we trim the data at 2.5 percent of the distribution. Note that we also obtain similar results even when we do not impose any size filters.

¹⁸ Throughout the paper, we calculate t -statistics using the [Newey and West \(1987\)](#) procedure with six lags.

Table 8

Univariate Portfolios of Swedish Equities Sorted by IVOL.

Each month from January 1988 to December 2018, stocks trading on the Swedish stock market are sorted into decile portfolios based on their IVOL. Panel A reports, for each decile, the average IVOL value, the next-month value-weighted average excess return, and the risk-adjusted returns (alphas) from the global AF4 models, respectively. The last column presents the 10-1 average IVOL, excess return, and risk-adjusted return spreads on the arbitrage portfolio with a long position in the portfolio of highest IVOL stocks (decile 10) and a short position in the portfolio of lowest IVOL stocks (decile 1). Panel B excludes the most illiquid 10 percent of stocks from the sample; the smallest 10 percent and the largest 5 percent of stocks from the sample, respectively. Newey-West adjusted *t*-statistics are reported in parentheses.

Panel A: Base Results											
	Low-IVOL	IVOL 2	IVOL 3	IVOL 4	IVOL 5	IVOL 6	IVOL 7	IVOL 8	IVOL 9	High-IVOL	High-Low
IVOL	0.0092	0.0127	0.0151	0.0173	0.0197	0.0224	0.0256	0.0300	0.0369	0.0543	0.0451 (50.54)
Excess Return	0.0089 (2.23)	0.0098 (2.45)	0.0100 (2.34)	0.0091 (2.09)	0.0100 (2.26)	0.0053 (1.01)	0.0038 (0.73)	0.0007 (0.11)	−0.0009 (−0.13)	−0.0201 (−2.45)	−0.0290 (−4.28)
AF4 Alpha	0.0035 (1.25)	0.0045 (1.63)	0.0037 (1.27)	0.0033 (1.42)	0.0053 (1.87)	0.0000 (−0.02)	0.0010 (0.28)	−0.0037 (−0.80)	−0.0046 (−0.90)	−0.0205 (−3.09)	−0.0241 (−4.01)
Panel B: Robustness Checks											
Excl. 10% Most Illiquid	Low-IVOL	IVOL 2	IVOL 3	IVOL 4	IVOL 5	IVOL 6	IVOL 7	IVOL 8	IVOL 9	High-IVOL	High-Low
AF4 Alpha	0.0050 (1.97)	0.0037 (1.40)	0.0047 (1.86)	0.0044 (1.69)	0.0039 (1.57)	0.0024 (0.81)	−0.0009 (−0.28)	0.0005 (0.10)	−0.0054 (−1.12)	−0.0147 (−2.37)	−0.0197 (−3.55)
Excl. 10% Smallest	Low-IVOL	IVOL 2	IVOL 3	IVOL 4	IVOL 5	IVOL 6	IVOL 7	IVOL 8	IVOL 9	High-IVOL	High-Low
AF4 Alpha	0.0043 (1.69)	0.0043 (1.65)	0.0043 (1.60)	0.0041 (1.65)	0.0031 (1.20)	0.0042 (1.45)	−0.0016 (−0.49)	0.0017 (0.36)	−0.0072 (−1.58)	−0.0154 (−2.53)	−0.0197 (−3.65)
Excl. 5% Largest	Low-IVOL	IVOL 2	IVOL 3	IVOL 4	IVOL 5	IVOL 6	IVOL 7	IVOL 8	IVOL 9	High-IVOL	High-Low
AF4 Alpha	0.0053 (1.96)	0.0039 (1.14)	0.0036 (1.26)	0.0029 (0.87)	0.0046 (1.40)	−0.0005 (−0.18)	0.0043 (1.23)	−0.0027 (−0.57)	−0.0059 (−1.20)	−0.0214 (−3.27)	−0.0267 (−4.78)

stocks) is positive but statistically insignificant (*t*-stat. = 1.25).¹⁹

Next, we analyze the strength of the IVOL puzzle across institutional ownership portfolios.²⁰ Specifically, at the end of each month *t*, all stocks in the sample are grouped into terciles based on an ascending sort of the level of institutional ownership and quintiles based on an ascending sort of the IVOL. Table 9 presents the AF4 alpha for each of the 15 (3 × 5) resulting intersection portfolios as well as the AF4 alpha differences between the extreme IVOL deciles along with the associated *t*-statistics. As shown in the last row of Table 9, the magnitude of the abnormal re-

Table 9

Bivariate Portfolios of Swedish Equities Sorted by Institutional Ownership and IVOL.

For each month from January 1999 to December 2007, stocks trading on the Swedish stock market are grouped into terciles based on an ascending sort of the institutional ownership levels (INST) for Swedish equities, and quintiles based on an ascending sort of the IVOL. The table presents the global AF4 alpha for each of the 15 (3 × 5) resulting intersection portfolios of INST and IVOL. The last row reports the AF4 alpha spreads for the hedge portfolio that is long in the decile of stocks with the highest IVOL and short in the quintile of stocks with the lowest IVOL for each of the tercile portfolio sorted by INST. The last column reports the difference between the AF4 alphas of the zero-cost IVOL portfolios among the extreme INST terciles (INST1 - INST3). Newey-West adjusted *t*-statistics are reported in parentheses.

	INST1	INST2	INST3	
Low-IVOL	0.0013 (0.26)	−0.0016 (−0.43)	0.0094 (2.66)	
IVOL 2	−0.0020 (−0.36)	0.0040 (1.09)	0.0042 (1.11)	
IVOL 3	−0.0170 (−2.62)	−0.0015 (−0.29)	0.0074 (1.61)	
IVOL 4	−0.0191 (−1.91)	0.0022 (0.34)	0.0093 (1.69)	
High-IVOL	−0.0264 (−2.79)	0.0022 (0.30)	0.0128 (1.25)	INST1-INST3
High - Low	−0.0277 (−3.13)	0.0038 (0.51)	0.0033 (0.32)	−0.0311 (−2.90)

¹⁹ We should note that our focus is on the one-month-ahead return predictability, following the asset pricing literature including widely-cited articles on idiosyncratic volatility and related stock market anomalies (e.g., Ang et al., 2006; Kumar, 2009; Bali et al., 2011). To provide robustness check, in Section B.1 of the online appendix, we investigate the six-month and 12-month-ahead return predictability using both Datastream (1988–2018) and Swedish household level data (1999–2007). Our results in Table O.A.18 show that the return predictability driven by idiosyncratic volatility is not just a one-month affair. The IVOL predicts cross-sectional variation in stock returns 12 months into the future. The anomalous negative relation between idiosyncratic risk and future returns is also confined to those stocks that are predominantly held by the wealthy households when we consider longer-term investment horizons.

²⁰ Note that we are not able to directly observe the institutional holdings at the security level. However, given that our household data cover the direct stockholdings of the entire Swedish population, we proxy for institutional holdings using the differences between the total number of shares outstanding of a firm and the total number of shares of that firm that are directly held by the Swedish population in a given year. We acknowledge that our institutional holdings measure may be imperfect. For example, foreign (rich) individual investors can also directly hold the stocks of companies in our sample, which we would consider in the holdings of the institutional investors. However, such a measurement issue would work against our findings, implying that our results may represent the lower bound effects of wealthy individuals on low risk anomalies.

turn to the zero-cost portfolio that buys stocks with the highest IVOL and sells stocks with the lowest IVOL decreases uniformly in absolute value as one moves towards the stocks for which the level of institutional holdings is highest (INST3). For those stocks in which retail investors are most active (INST1), we observe that stocks in the highest IVOL quintile lose a one-month-ahead AF4 alpha

of 2.64% per month. The AF4 alpha spread is economically large and statistically significant; -2.77% (t -stat. = -3.13). The difference between the AF4 alpha of the zero-cost IVOL portfolios among the extreme institutional ownership terciles (INST1 - INST3) is also highly significant; -3.11% (t -stat. = -2.90). Overall, consistent with the prior literature (Kumar, 2009; Han and Kumar, 2013), these results indicate that the IVOL puzzle is primarily driven by stocks that are largely held by individual investors.²¹

4.2. Rich households and the IVOL puzzle

The high concentration of direct stock wealth in the hands of the rich combined with our micro-level evidence in Section 3 that richer households have significantly higher exposure to idiosyncratic stock risk imply that it makes sense to focus on the portfolio selection behavior of the rich households (and their preferences) if one wants to understand the impact of individual investors on equilibrium stock prices, and their potential contributions to the asset pricing puzzles, such as the low risk anomalies. In what follows, we turn to examine this issue.

4.2.1. Bivariate portfolio-level analysis

To investigate the relation between the IVOL puzzle and the demand of rich households, we first conduct a bivariate portfolio-level analysis. Specifically, we use the stock-level portfolio information of the wealthy households, and construct a variable, RICH, that measures the ownership level of rich individual investors in a given stock. Specifically, our measure takes the following form:

$$RICH_{i,t} = \frac{Shares_{i,t}^W}{Shares_Outstanding_{i,t}} \quad (3)$$

where $RICH_{i,t}$ is the share of company i 's outstanding stocks that are directly held by the wealthiest households who are identified as those in the top 10 percent of the wealth distribution in year t . Analogously, $Shares_{i,t}^W$ is the total number of shares of company i that are directly held by the wealthy households in year t , and $Shares_Outstanding_{i,t}$ is the total number of shares outstanding of company i at the end of year t .

As shown in Panel A of Table O.A.21, the wealthy households directly own approximately 16 percent of the average publicly traded company in Sweden.²² In Panel B of Table

O.A.21, we report pairwise correlations between the individual stock characteristics and the rich ownership variable. Importantly, there is a positive and significant correlation between the demand of rich investors, as measured by the RICH variable, and the IVOL of stocks ($\rho = 0.25$ with a p -value < 0.01).

Next, we group all stocks into five quintile portfolios based on an ascending sort of the rich ownership (i.e., RICH) variable at the end of each month. We then sort all stocks in each wealth quintile into five quintile portfolios based on an ascending sort of their IVOL, which results in 25 (5×5) value-weighted portfolios. Panel A of Table 10 reports next-month average excess return, and the AF4 alpha spreads between the high-IVOL and low-IVOL quintile portfolios resulting from the 25 intersection portfolios along with their corresponding t -statistics. We consider the stock investments of households who are in the top 10 percent of the wealth distribution when defining the RICH variable.

Consistent with the results from the household level analysis, we find that the alpha spreads between the high- and low-IVOL stocks are negative and economically and statistically significant for those stocks that are widely held by rich individuals. For example, based on the estimates in Panel A, the AF4 alpha of the zero-cost portfolio that buys stocks with the highest IVOL and sells stocks with the lowest IVOL in the highest rich ownership quintile accounts for -2.48% (t -stat. = -2.31) per month. Interestingly, they are statistically insignificant in the lower quintiles with lower levels of wealthy investor ownership. These results highlight an interesting pattern. The anomalous negative relation between idiosyncratic risk and future returns is only confined to those stocks that are predominantly held by the wealthy individuals, whereas there is no such effect for stocks with lower levels of rich investor ownership.

A powerful alternative interpretation of our results is that rich investor ownership may proxy for overall individual investor demand. Under this interpretation, our findings may simply be a (re-)manifestation of the well-known relationship between individual ownership and the low risk anomalies (Kumar, 2009; Han and Kumar, 2013). We address this concern in several ways. First, we restrict our sample to the stock investments of households in the bottom 80 percent of the wealth distribution, and construct a stock-level measure that captures the share of a given company's outstanding stocks that are directly held by the non-wealthy households. The results of the bivariate sort analysis are presented in Panel B of Table 10. Interestingly, the risk-adjusted returns of the arbitrage portfolios (High-Low IVOL) across all quintiles turn out to be statistically insignificant, once we focus on the ownership of non-wealthy individuals. Second, we employ an alternative measure, analogous to Eq. (3), for rich investor ownership where we eliminate the total holdings of the households in the bottom 90 percent of the wealth distribution both from the total household holdings and total shares outstanding of a given company. The results are reported in Panel C of Table 10. Even after removing the holdings of the non-wealthy households, we still find both statistically and economically significant negative AF4 alpha spread, -3.61% per month (t -stat. = -3.14), among

²¹ In Tables O.A.19–O.A.20 of the online appendix, we also investigate the mispricing explanation of the IVOL puzzle by proposing a stock-level mispricing measure (MISP) for Sweden, which is in the spirit of the measure of Stambaugh et al. (2012, 2014, 2015), and perform a bivariate portfolio analysis by sorting the stocks according to their MISP and IVOL scores. As discussed in Section B.2 of the online appendix, our results indicate that the IVOL puzzle is much stronger for overpriced stocks and support the mispricing hypothesis that the high-IVOL stocks are subject to overpricing, suggesting that the low risk anomaly is driven by retail investors' demand for high-volatility stocks, leading to temporary overpricing and negative future abnormal returns for these high-volatility stocks with large lottery payoffs.

²² We note that this figure represents the lower bound of the rich ownership rate because (wealthy) individuals can also own companies indirectly through mutual funds or trusts. We do not consider their investments through professional asset management companies, as we intend to solely focus on the investment behavior of the individual investors in our analysis.

Table 10

Bivariate Portfolios of Swedish Stocks Sorted by IVOL and Rich Investor Ownership.

For each month from January 1999 to December 2007, stocks trading on the Swedish stock market are first grouped into quintiles based on an ascending sort of the rich investor ownership variable (RICH). Then, within each RICH quintile portfolio, stocks are further sorted into subquintiles based on an ascending sort of the IVOL. We present the global AF4 alpha spreads for the hedge portfolio that is long in the quintile of stocks with the highest IVOL and short in the quintile of stocks with the lowest IVOL for each of the quintile portfolio sorted by RICH. In Panel A, we present the results of the base analysis. In Panel B, we consider the stock investments of households who are in the bottom 80 percent of the wealth distribution in a given year when defining the non-RICH variable. In Panel C, we use an alternative definition of the RICH variable (i.e., $RICH^*$) in which we eliminate the total holdings of households in the bottom 90 percent of the wealth distribution from both the total household holdings and the total shares outstanding of a given firm. In Panel D, we use an orthogonalized $RICH$ variable that is net of the total retail holdings in a given stock in a given month. Newey-West adjusted t -statistics are reported in parentheses.

Panel A: Base Results					
Top 10%	Low-RICH	RICH 2	RICH 3	RICH 4	High-RICH
Average returns	−0.0098 (−1.30)	−0.0108 (−0.81)	−0.0224 (−1.54)	−0.0409 (−3.51)	−0.0367 (−3.31)
AF4 Alpha	−0.0086 (−1.47)	−0.0035 (−0.36)	−0.0152 (−1.28)	−0.0274 (−3.87)	−0.0248 (−2.31)
Panel B: Investors Outside the Top of the Wealth Distribution					
Bottom 80%	Low-Non-RICH	Non-RICH 2	Non-RICH 3	Non-RICH 4	High-Non-RICH
AF4 Alpha	0.0017 (0.19)	0.0041 (0.46)	0.0027 (0.27)	−0.0185 (−1.49)	−0.0168 (−1.45)
Panel C: Alternative Definition of the RICH Measure ($RICH^*$)					
Top 10%	Low- $RICH^*$	$RICH^*$ 2	$RICH^*$ 3	$RICH^*$ 4	High- $RICH^*$
AF4 Alpha	−0.0115 (−1.76)	0.0015 (0.16)	−0.0081 (−0.77)	−0.0318 (−2.95)	−0.0361 (−3.14)
Panel D: Orthogonalized RICH Measure ($\widehat{RICH}$)					
Top 10%	Low- $\widehat{RICH}$	$\widehat{RICH}$ 2	$\widehat{RICH}$ 3	$\widehat{RICH}$ 4	High- $\widehat{RICH}$
AF4 Alpha	−0.0086 (−1.48)	−0.0034 (−0.36)	−0.0152 (−1.28)	−0.0274 (−3.87)	−0.0248 (−2.31)

stocks with higher wealthy investor ownership. Finally, we also focus on an orthogonalized RICH measure, which corresponds to the residual holdings of wealthy households in a given stock with respect to the average holdings of all households in the sample in a given period. As shown in Panel D of Table 10, the AF4 alpha spread of the zero-cost portfolio in the highest rich ownership quintile retains its economic and statistical significance; it accounts for −2.48% per month with a t -statistic of −2.31.

To ensure the robustness of our findings, we perform several additional tests. First, the Swedish stock market is dominated by a small number of large companies (e.g., Ericsson, Volvo), which are also widely held by a majority of stockholding households (Betermier et al., 2017). To address this issue, we exclude from the sample the five most popular stocks held by Swedish households and repeat our empirical analyses. As reported in Panel A of Table O.A.22, we obtain similar results, indicating that our results are not driven by popular stocks. Second, to alleviate any concerns that large- or micro-cap stocks may drive our results, we eliminate those stocks that are in the top 1% or bottom 1% of the size distribution based on the market value and perform the bivariate portfolio analysis. The results are presented in Panels B and C of Table, respectively. Our findings are robust to the exclusion of micro- or large-cap stocks from the sample. Third, to address potential concerns that our results may be driven by the boom and bust of the tech bubble in the early sample period, we next restrict our attention to the period between 2002 and 2007

and check the sensitivity of our findings to the exclusion of the internet bubble period. As presented in Panel D of Table O.A.22, we obtain similar results. Fourth, to ensure that our results are not driven by any outliers, we winsorize the return variable at the 2.5% upper and lower tails of the distribution. Panel E of Table O.A.22 reports the results of this sensitivity check. We find that the monthly alpha spreads between the high- and low-IVOL stocks with the highest rich ownership retain their economic and statistical significance. Finally, we consider other measures of low risk anomalies such as beta, value-at-risk, and expected shortfall. As presented in Table O.A.23 of the online appendix, we again find that the anomalous relation between risk and return is mainly observed among those stocks that are predominantly held by rich households.²³

4.2.2. Switcher analysis

The analysis in the previous section shows that there is a robust positive association between high-IVOL stocks and

²³ Since we are not able to observe households' intra-year stock holdings and trades, in the analysis so far we have constructed the RICH measure based on households' year-end holdings, assuming that (rich) households hold the same stock portfolio within a given reporting year. An alternative approach is to construct the RICH measure on the basis of holdings at the beginning of the year, assuming that households hold the same stock portfolio until the last day of the following reporting year. As reported in Tables O.A.24–O.A.27 of the online appendix, we obtain very similar results when we use the RICH measure based on household holdings at the beginning of the year.

the aggregate demand of rich investors. However, we recognize that a causal interpretation of the results from the bivariate portfolio-level analysis is subject to some limitations due to identification issues.²⁴ To provide a more direct, causal interpretation of our results, we next conduct multivariate regressions where we exploit exogenous variation in the aggregate demand of the rich households (i.e., the RICH variable) for individual stocks generated by a rule exception in the inheritance and gift taxation of household stock wealth.

First, it is important to note that inherited wealth has a substantial share in private wealth in Sweden, particularly among the richer households (Nekoei and Seim, 2023). For example, Ohlsson et al. (2020) estimate this share to be as high as 80 percent in the early 1900s, and to be approximately 50 percent since the 1950s. Accordingly, there was an inheritance and gift tax in Sweden that was introduced in 1885 and abolished in 2004 (Rietz et al., 2015). The tax rate varied substantially over this period, ranging from 0.5% in the early periods to a maximum of 70% in 1983. During our sample period, the statutory top marginal rate accounted for 30% (Rietz et al., 2015), but there were some exemptions in the valuation of inheritances and *inter vivos* across different asset classes. One such exemption, that is relevant for us, was the difference in the inheritance and gift taxation of stock investments in large-cap (i.e., A-listing) and mid-/small-cap (O-listings) stocks. Specifically, large-cap stocks were valued at 75% of the stock price, while mid-/small-cap stocks were valued at 30% of their price at the time of assessment, which generated a large asymmetry in the inheritance and gift taxation of stocks listed on different market segments (Jennergren, 2004). Given this background, if a stock moves from the A-list to the O-list, it creates tax incentives among richer households, who are more prone to changes in inheritance and gift taxation, to rebalance their investments towards these stocks. Hence, the change in taxation status would create reasonably exogenous variation in the RICH measure among switcher stocks, which allows us to pin down the effect of demand of the rich on the changes in the IVOL of a stock.

To test this idea, we collect auxiliary information from Statistics Sweden about the market segment of listed stocks for each year between 1999 and 2004.²⁵ We then identify those stocks that were initially A-listed (and subject to a higher taxation), but changed their market segment to the O-list and became subject to lower taxation during the observation period. We label these stocks as switcher stocks. We identified 7 switcher stocks (out of 69 stocks that were initially A-listed) over the sample period.

Importantly, narratives make clear that the dominant factor for these switcher companies to move from the A-list to the O-list was indeed tax reasons related to inher-

itances and *inter vivos*, rather than reasons related to the fundamentals (Jennergren, 2004).²⁶ Accordingly, as presented in Panel A of Table O.A.28, we observe that the RICH measure is significantly larger among stocks listed on the O-list as compared to those listed on the A-list. We recognize that this difference could simply be due to the smaller market capitalization or wealth tax exemption status of O-listed stocks as compared to that of the A-listed stocks.²⁷ More convincing support, presented in Panel B of Table O.A.28, comes from the sample of switcher stocks: when a stock moves from the A-list to the O-list, we observe a significant increase in the demand of the rich investors for that stock around the change in its taxation status.

To provide a more formal analysis on the effect of RICH on IVOL, we next rely on a difference-in-difference type design and exploit the exogenous changes in the RICH variable of a stock generated by changes in its taxation status. Specifically, we estimate the following multivariate regressions:

$$IVOL_{imt} = \beta_1 \cdot RICH_{imt} + \beta_2 \cdot Post_Tax_Change_{imt} + \beta_3 \cdot RICH_{imt} \times Post_Tax_Change_{imt} + \beta_4 \cdot Stock_Controls_{imt} + \gamma_i + \gamma_t + \epsilon_{imt} \quad (4)$$

where $Post_Tax_Change_{imt}$ is an indicator variable that takes the value of 1 if stock i is listed on the O-list (thus, is subject to lower inheritance and gift taxation) in month m in year t , and 0 otherwise. In these regressions, we also account for stock fixed effects (γ_i) and time fixed effects (γ_t) along with some time-varying controls, denoted by vector $Stock_Controls_{imt}$, such as price, market capitalization, beta, trading volume, and momentum of stock i in month m in year t . We correct standard errors by clustering at the time-dimension. Note that our sample also includes A-listed stocks that did not change their market segment throughout the observation period along with the switcher stocks. To minimize the potential effects of other confounding factors, we eliminate observations in the treatment year for the switcher stocks.

The regression results are reported columns (1) and (2) of Table 11. We indeed document that the increases in the aggregate demand of rich households for a particular

²⁴ For example, any (time-invariant or time-varying) unobserved stock characteristics can simultaneously affect the demand of the rich households and the IVOL of a stock, yielding a positive correlation between the two variables without implying any causal link.

²⁵ Note that we use stock data up until 2004 in this analysis, as the inheritance and gift taxation was repealed by the Swedish government in late 2004.

²⁶ See, for example, Jennergren (2004) who writes that companies that moved from the A-list to the O-list between 2000 and 2003 cite tax reasons, referring to the inheritance and gift taxes. Similarly, the press releases of the companies that changed their listing during our observation period show that the main motivation for these companies was to generate some benefits for their shareholders by reducing the tax burden of investing in the shares of these companies (see, for example, <https://www.globenewswire.com/news-release/2002/11/14/292030/1128/en/The-Axfood-Share-to-Move-from-the-A-list-to-the-O-list.html>). Finally, it is also important to note that there were no companies that moved from the O-list to the A-list during the sample period.

²⁷ All residents in Sweden (with net wealth above a certain threshold) were subject to taxation of their wealth over the sample period. Household investments in the O-listed stocks were exempt from wealth taxation, while wealth invested in the A-listed stocks were subject to wealth taxation. However, changes in the wealth tax law (Prop. 1997/98:1) state that all stocks that moved from the A-list to the O-list after May 1997 would be still subject to wealth taxation (Jennergren, 2004). Hence, our analysis based on switcher stocks is immune to concerns related to changes in wealth taxation of these stocks.

Table 11

Switcher Analysis: IVOL Puzzle.

This table presents coefficient estimates from panel regressions for IVOL and (one-month-ahead) risk-adjusted returns, as measured by AF4, at the stock level for the period between 1999–2004. In columns (1) and (2), stock level IVOL is regressed on the share of outstanding stocks held by households in the top 10 percent of the wealth distribution, an indicator variable for changes in taxation status of the stock, and other stock characteristics. In columns (3) and (4), (one-month-ahead) AF4 is regressed on the IVOL, an indicator variable for changes in the tax status of the stock, and other stock characteristics. All regressions account for time (year) and stock fixed effects. Standard errors are clustered at the year-month level. The corresponding *t*-statistics are reported in parentheses. See Data Appendix for detailed variable definitions.

	IVOL		AF4 _{t+1}	
	(1)	(2)	(3)	(4)
Post_Tax_Change	−0.0032 (−2.20)	−0.0031 (−2.13)	0.0133 (4.41)	0.0151 (5.08)
Post_Tax_Change × RICH	0.0201 (2.72)	0.0213 (2.92)		
Post_Tax_Change × IVOL			−0.2369 (−1.93)	−0.2686 (−2.22)
Beta		0.0049 (2.97)		0.0023 (1.30)
Volume		0.0002 (2.46)		0.0004 (4.09)
Price		−0.00001 (−0.30)		−0.0001 (−1.72)
Momentum		−0.0019 (−2.58)		0.0078 (11.25)
MV		−0.0003 (−1.82)		0.0016 (8.56)
RICH	0.0169 (1.87)	0.0111 (1.11)		
IVOL			−0.0039 (−0.09)	0.0512 (1.11)
Constant	0.0176 (18.48)	0.0159 (10.93)	0.0090 (10.01)	0.0013 (1.25)
No of Obs	3632	3580	3454	3405
FE: stock	Yes	Yes	Yes	Yes
FE: time	Yes	Yes	Yes	Yes

stock have a significant positive effect on the IVOL of a stock when the stock is subject to lower taxation, with *t*-statistics ranging from 2.72 to 2.92 depending on the regression specification. Note that these results are robust to the inclusion of stock and year fixed effects, and other time-varying stock characteristics.

Next, to examine whether the documented effects extend beyond the IVOL to the valuation of stocks, we use one-month-ahead risk-adjusted stock returns, as measured by the AF4 alpha, as the outcome variable. Specifically, we estimate the following regression:

$$AF4_{imt+1} = \beta_1 \cdot IVOL_{imt} + \beta_2 \cdot Post_Tax_Change_{imt} + \beta_3 \cdot IVOL_{imt} \times Post_Tax_Change_{imt} + \beta_4 \cdot Stock_Controls_{imt} + \gamma_i + \gamma_t + \epsilon_{imt} \quad (5)$$

The results are reported in columns (3) and (4) of Table 11. Consistent with the idea that the rich demand partly contributes to the IVOL puzzle, we indeed find that the IVOL of the stock has a negative and statistically highly significant effect on the stock's next month's alpha for those firms that switched from A- to O-listing in the period following the switch (*t*-stat. = −2.22). In additional tests, presented in Table, we augment the regression specification in Eq. (5) with a triple interaction term between IVOL, *Post_Tax_Change*, and the RICH variable, and find that the coefficient estimate on the triple interaction term is negative and statistically significant (*t*-stat. = −3.96). Taken together, these additional empirical results imply that an in-

crease in rich households' aggregate demand for a particular stock not only leads to an increase in the IVOL of that stock, but also to significantly more negative risk-adjusted returns in the following month.

Overall, multivariate regressions provide further support for the causal interpretation that it is the rich demand that partly drives the IVOL puzzle.

4.2.3. Status-driven lottery demand factor

We finally investigate the role of rich households in the existence of the IVOL puzzle by constructing a pricing factor, which builds on the model predictions of Becker et al. (2000) and Roussanov (2010) and our household level evidence in Section 3. First, we test at the stock level whether stocks that are primarily held by rich households display highly idiosyncratic, lottery-like payoffs. To do so, we form univariate portfolios for each month from January 1999 to December 2007 by sorting all stocks into quintiles based on their rich ownership levels. Table 12 presents the median characteristics of stocks by rich household ownership. We observe that stocks that are predominantly held by wealthy households are significantly smaller, less liquid, and lower-priced which are typical characteristics of high-IVOL stocks with strong lottery features (see, e.g., Kumar (2009); Han and Kumar (2013), and Bali et al. (2011, 2017)). In fact, Table 12 shows that stocks that are largely held by rich households have higher idiosyncratic volatility (IVOL) and stronger lottery payoffs (MAX), suggesting that rich individuals tend to

Table 12

Median Lottery Characteristics of Swedish Equities Sorted by Rich Investor Ownership.

Each month from January 1999 to December 2007, stocks trading on the Swedish stock market are sorted into quintile portfolios based on their level of rich investor ownership (RICH). This table reports, for each quintile, the median values of the rich ownership level (RICH), maximum daily returns (MAX), idiosyncratic volatility (IVOL), market capitalization (SIZE), price (PRICE), and illiquidity (ILLIQ). The last column presents the 5–1 differences on the arbitrage portfolio with a long position in the portfolio of highest RICH stocks (quintile 5) and a short position in the portfolio of lowest RICH stocks (quintile 1). T-statistics are reported in parentheses.

	Low-RICH	RICH 2	RICH 3	RICH 4	High-RICH	High-Low
RICH	0.0376	0.0800	0.1248	0.1899	0.3136	0.276 (86.42)
IVOL	0.0181	0.0197	0.0218	0.0241	0.0305	0.0124 (22.02)
MAX	0.0267	0.0289	0.0309	0.0340	0.0416	0.0149 (16.45)
SIZE	6.4691	5.6852	4.5840	4.0185	2.9904	−3.4787 (−75.06)
PRICE	7.6568	6.7447	4.5879	4.4451	3.5103	−4.1465 (−23.62)
ILLIQ	0.0001	0.0001	0.0007	0.0012	0.0034	0.0033 (7.64)

tilt their portfolios to high-IVOL stocks with strong lottery characteristics. Consistent with the results presented in Tables 2 to 6, and also in line with the model of Roussanov (2010), Table 12 shows that stocks that are widely held by rich investors display high volatility, and the demand of rich investors for stocks with high idiosyncratic risk can be attributed to their preferences for lottery-like features.

Next, we introduce a status-driven lottery demand factor and test if this newly proposed factor produces significant return and alpha spreads. We form a household-level equity factor, denoted RICHMAX, using the factor-forming technique pioneered by Fama and French (1993). At the end of each month t , we sort all Swedish stocks into five portfolios based on the ownership level of rich individual investors in a given stock (RICH), proxying for the level of social status based on the theoretical model of Roussanov (2010). We independently sort all Swedish stocks into five portfolios based on the average of five highest daily returns over the past month (MAX), proxying for lottery preference à la Bali et al. (2011, 2017). The intersections of the five RICH-sorted portfolios and the five MAX-sorted portfolios generate a total of 25 (5×5) portfolios. The status-driven lottery demand factor (RICHMAX) return in month $t + 1$ is taken to be the average return of the value-weighted high-MAX and high-RICH portfolio minus the average return of the value-weighted low-MAX and low-RICH portfolio. As such, the RICHMAX factor portfolio is designed to capture returns associated with retail investors' preference for lottery-like stocks while accounting for the difference between their social status concerns. The RICHMAX factor generates an average monthly return and an AF4 alpha of -4.16% (t -stat. $=-3.27$) and -2.42% (t -stat. $=-2.42$), respectively.²⁸

²⁸ We also construct an alternative status-driven lottery demand factor using the holdings of both rich households and lottery bond investors (as observed in the Swedish Wealth and Income Registry). Specifically, we independently sort all Swedish stocks into five portfolios based on the holdings of lottery bond investors. The intersections of the five RICH-sorted portfolios and the five LOTTERY-sorted portfolios yield a total of 25 (5×5) portfolios. The return of the alternative status-driven lottery demand factor (RICH_LOTTERY) in month $t + 1$ is taken to be the aver-

Finally, we test if the status-driven lottery demand factor (RICHMAX) explains the idiosyncratic volatility puzzle in Sweden. We first estimate the AF4 alpha spread (high-IVOL minus low-IVOL) on the value-weighted univariate portfolios of idiosyncratic volatility; -2.45% per month (t -stat. $=-2.78$). Then, we extend the AF factor model by including the RICHMAX factor and estimate the extended factor model alpha on IVOL-sorted portfolios. The (AF4+RICHMAX) alpha reduces to -1.28% per month and becomes statistically insignificant with a t -statistic of -1.67 . These results indicate that social status concerns of rich households and the induced preferences for lottery-like securities explain rich investors' demand for high-risk stocks, resulting in overpricing and low future returns for such stocks.²⁹

5. International evidence

The rich nature of the Swedish household level data provides a unique opportunity to test the relationship between wealthy investor demand and the volatility puzzle. At the same time, though, it may raise concerns that our findings are specific to Sweden. To provide external

age return of the value-weighted high-LOTTERY and high-RICH portfolios minus the average return of the value-weighted low-LOTTERY and low-RICH portfolios. We find that the RICH_LOTTERY generates an average monthly return and AF4 alpha of -1.41% (t -stat. $=-1.97$) and -1.15% (t -stat. $=-1.69$), respectively.

²⁹ To measure the relative contribution of RICH- and MAX-sorted portfolios to the RICHMAX factor, we work with the 5×5 independent bivariate portfolios of stocks sorted by RICH and MAX. Panel A of Table O.A.30 of the online appendix shows that the average return, CAPM and AF4 alpha spreads on MAX-sorted portfolios averaged across the RICH quintile are all negative, in the range of -1.55% to -2.80% per month, and statistically significant with t -statistics ranging from -2.21 to -4.57 . Panel B of Table O.A.30 in the online appendix shows that the average return, CAPM and AF4 alpha spreads on RICH-sorted portfolios averaged across the MAX quintile are all negative but smaller in absolute magnitude, in the range of -0.68% to -1.16% per month, and statistically weaker with t -statistics ranging from -1.44 to -1.89 . These results indicate that although the wealth effect combined with the lottery-type preferences of individual investors magnifies the strength of the RICHMAX factor, preference for lottery stocks of retail investors contributes more than their wealth to the performance of the RICHMAX factor.

validity, this section investigates whether the puzzling, negative relation between idiosyncratic volatility and future equity returns is significant in the cross-section of stocks trading in the U.S. and international equity markets. For the international analysis, we focus on 23 developed countries used by [Fama and French \(2012, 2017\)](#). These countries can be grouped into four regions: North America, including the United States and Canada; Europe, including Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom; Japan; and Asia-Pacific, including Australia, Hong Kong, New Zealand, and Singapore. The international equity return data are obtained from Datastream and cover the period from January 1988 to December 2018. This sample includes 3.2 million firm-month observations for which the key variable of interest, idiosyncratic volatility, can be calculated.³⁰

5.1. The low-risk anomaly in the U.S. and international countries

We begin our analysis by examining whether the low risk anomaly is significant in the cross-section of individual stocks trading in the U.S. and international markets. We first present evidence for the U.S. equity market and then for the remaining 22 developed countries. For each month from July 1963 to December 2018, stocks trading in the U.S. equity market are sorted into decile portfolios based on their idiosyncratic volatility (IVOL), where decile 1 (decile 10) contains stocks with the lowest (highest) IVOL. Following earlier studies on idiosyncratic volatility for the U.S. equity market (e.g., [Ang et al., 2006](#)), we calculate the one-month-ahead value-weighted average portfolio returns, and this procedure is repeated each month until the sample is exhausted.

Table O.A.31 of the online appendix reports for each decile the average idiosyncratic volatility, the next-month average excess return, and the risk-adjusted returns (alphas) based on the four-factor model (FFC) of [Fama and French \(1993\)](#) and [Carhart \(1997\)](#), and the seven-factor model (FF6PS) of [Fama and French \(1993, 2015, 2018\)](#), [Carhart \(1997\)](#), and [Pástor and Stambaugh \(2003\)](#). The last column of Table O.A.31 presents the 10-1 average IVOL, excess return, and risk-adjusted return (alpha) spreads for the arbitrage portfolio that is long in the decile of stocks with the highest IVOL and short in the decile of stocks with the lowest IVOL. Consistent with [Ang et al. \(2006\)](#), the univariate portfolio sorts of U.S. equities indicate a significantly negative relation between IVOL and next-month average returns. The arbitrage portfolio (High-Low IVOL) with a long position in the high-volatility stocks and a short position in the low-volatility stocks loses on average 0.60% per month (or 7.2% per annum) with a Newey-West *t*-statistic of 2.38. Next, we examine whether the excess return differences between the IVOL deciles can be explained by standard asset pricing models. The last two rows in Table O.A.31 present the next month's risk-adjusted returns

for the High-Low IVOL portfolio. The FFC and FF6PS alpha spreads for the long-short portfolio are negative, economically large and highly significant. Thus, we conclude that the low risk anomaly is not explained by the market, size, value, momentum, liquidity risk, investment, and profitability factors of [Fama and French \(1993, 2015, 2018\)](#), [Carhart \(1997\)](#), and [Pástor and Stambaugh \(2003\)](#).

Finally, we investigate the source of the economically large risk-adjusted return spreads between the high- and low-IVOL portfolios. For this, we focus on the economic and statistical significance of the risk-adjusted returns of decile 1 versus decile 10 of the value-weighted portfolios. As reported in Table O.A.31, the alphas of decile 1 (low-volatility stocks) are economically small, and statistically insignificant, whereas the corresponding alphas of decile 10 (high-volatility stocks) are negative, economically large, and highly significant. Thus, we conclude that the significantly negative alpha spread between the high-IVOL and low-IVOL stocks is driven by underperformance of stocks in the high-IVOL decile. Overall, these results are consistent with prior theoretical and empirical evidence that investors' preference for high-volatility stocks induce them to pay high prices for these stocks, leading to temporary overpricing and negative future abnormal returns for such stocks. This explains the economically large and statistically significant negative alpha on high-IVOL stocks, generating the low risk anomaly in the cross-section of U.S. equities.

Panels A, B, and C of Table O.A.32 of the online appendix present the value-weighted abnormal return comparisons between equity deciles formed based on IVOL for various country groupings. In this analysis, we combine all equities in each subsample, thus, stocks in each decile may come from a multitude of countries. The risk-adjusted returns (alphas) on IVOL-sorted portfolios are estimated using the global [Asness and Frazzini \(2013\)](#) (AF4 alpha) model. Panel A of Table O.A.32 presents results for the full international sample (22 developed countries, excluding the US). Similar to our findings for the U.S. equities, the average monthly return and alpha spreads are negative and significant for developed countries; -1.58% (t -stat. = -3.25) for excess returns and -1.43% (t -stat. = -4.55) for the AF4 model. Next, we focus on two subsamples among developed markets, namely G10 and G7 countries (excluding the US). Leaving the U.S. aside, countries included in the G10 are Belgium, Canada, France, Germany, Italy, Japan, Netherlands, Sweden, Switzerland, and the United Kingdom, whereas Belgium, Netherlands, Sweden, and Switzerland are not included in the G7. The results for the univariate portfolios for these two subsamples are reported in Panels B and C of Table O.A.32. For both G10 and G7 countries, the long-short portfolio has economically large and highly significant return and alpha spreads. Another notable point in Panels A, B, and C is that the idiosyncratic volatility puzzle in developed markets as well as in G10 and G7 countries is again driven by the underperformance of high-volatility stocks, as in the U.S., with decile 10 having significantly negative abnormal returns (alphas), whereas decile 1 has economically and statistically insignificant alphas.

³⁰ Section B.3 of the online appendix provides a detailed description of the dataset for U.S. and international countries.

5.2. Does investors' demand for lottery stocks explain the low-risk anomaly?

In this section, we test whether investors' demand for lottery-like stocks explain the idiosyncratic volatility puzzle in the U.S. and international markets. We proceed by generating a factor capturing the returns associated with lottery demand and examining the ability of this factor to explain the returns associated with the volatility puzzle. Following Bali et al. (2017), we form a lottery-demand factor, denoted FMAX, following Fama and French (1993). Panel A of Table O.A.33 of the online appendix presents the abnormal returns for the IVOL-sorted decile portfolios of U.S. equities using the FFC+FMAX and FF6PS+FMAX factor models. Compared to our results reported in Table O.A.31, inclusion of FMAX in the factor models has dramatic effects on the abnormal returns. When the FFC model is augmented with the FMAX factor (FFC+FMAX), the alpha of the High-Low IVOL portfolio of -0.20% per month is both economically small and statistically indistinguishable from zero, with a t -statistic of -1.95 . This compares to an alpha of -0.78% per month (t -stat. = -5.30) relative to the FFC model (Table O.A.31). A similar result holds when the FF6PS factor model is extended with the FMAX factor. Using the FF6PS+FMAX model, the High-Low IVOL portfolio's alpha is -0.17% per month with a t -statistic of -1.58 , compared to -0.42% per month with a t -statistic of -3.09 using the FF6PS model (Table O.A.31).

Next, we construct the FMAX factor using individual stocks trading in 22 developed, G10, and G7 countries. Panels B, C, and D of Table O.A.33 of the online appendix present the AF4+FMAX alpha spreads on IVOL-sorted decile portfolios of stocks trading in international markets. The results indicate that when the lottery demand factor is included in the asset pricing models, the alpha of the High-Low IVOL portfolio becomes economically smaller and statistically insignificant for all three samples constructed with stocks trading in 22 developed, G10, or G7 countries. Overall, the results from the U.S. and international equity markets indicate that investors' demand for lottery-like stocks is an important driver of the idiosyncratic volatility puzzle. The volatility anomaly is no longer detected when IVOL-sorted portfolios are neutralized to lottery demand.

To provide further evidence on the lottery demand based explanation of the volatility puzzle, we calculate the cross-sectional correlation between idiosyncratic volatility (IVOL) and the lottery characteristic (MAX) of individual stocks trading in the US, 22 developed, G10, and G7 countries. Figures O.A.1–O.A.4 of the online appendix demonstrate the cross-sectional correlations between IVOL and MAX for each month in respective sample. Figure O.A.1 of the online appendix shows that the correlation between IVOL and MAX for the U.S. equity market fluctuates between 0.80 and 0.96, with an average correlation of 0.88. Figures O.A.2–O.A.4 present very similar correlations for international equity markets; the average correlation between IVOL and MAX is 0.85, 0.86, and 0.87 for 22 developed, G10, and G7 countries, respectively. Overall, the results presented in Table O.A.33 and Figures O.A.1–O.A.4 show that the low-risk anomaly exists only when the price

impact of lottery demand falls disproportionately on high-IVOL stocks, indicating that investors' preference for high-volatility stocks with large lottery-like payoffs induce them to pay high prices for these stocks, leading to temporary overpricing and negative future abnormal returns for such stocks.

5.3. Retail investors' demand for lottery stocks and the low-risk anomaly

In this section, we investigate whether the magnitude of the negative relation between idiosyncratic volatility and future returns is larger for those stocks in which individual investors are more active compared to those stocks in which institutional investors are more active. We test this hypothesis both in the U.S. and international equity markets.

We analyze the strength of the low-risk anomaly across institutional ownership portfolios using a bivariate sort analysis. At the end of each month t , we first sort stocks into terciles based on the level of institutional ownership (INST). Then, within each INST tercile, we further sort stocks into 10 decile portfolios based on idiosyncratic volatility (IVOL) to generate 3×10 portfolios of INST and IVOL. We form an arbitrage portfolio that is long in the resulting high-IVOL portfolio and short in the resulting low-IVOL portfolio within each INST tercile. Table O.A.34 of the online appendix presents the FF6PS alpha for each of the 30 (3×10) resulting portfolios as well as the FF6PS alpha differences between the extreme IVOL deciles along with the associated t -statistics for the U.S. equity market. As shown in the last row of Table O.A.34, the magnitude of the abnormal return to the zero-cost portfolio that buys stocks with the highest IVOL and sells stocks with the lowest IVOL increases uniformly in absolute value as one moves towards the stocks for which the level of institutional holdings is lowest (INST1). For those stocks in which retail investors are most active, the FF6PS alpha spread is economically large and statistically significant; -2.04% per month (t -stat. = -7.81). This alpha spread is almost four times as high in absolute magnitude with respect to that observed in the highest institutional ownership tercile. The difference between the FF6PS alphas of the zero-cost IVOL portfolios among the extreme institutional ownership terciles (INST1 - INST3) is also highly significant; -1.56% per month (t -stat. = -3.77). Collectively, these results indicate that the IVOL effect is much stronger for equities largely held by retail investors.

Next, we try to replicate these analysis using individual stocks trading in 22 developed, G10, and G7 countries. However, we do not have institutional (or retail) ownership of individual stocks trading in international markets. Thus, we proxy INST with firm size as institutional (retail) investors are known to hold relatively large/liquid (small/illiquid) stocks. That is, we form 3×10 bivariate portfolios of INST and IVOL by replacing INST with firm size (market cap) for 22 developed countries. Table O.A.35 of the online appendix presents very similar results for international equity markets. For example, the alpha spread on IVOL-sorted portfolios of stocks trading in 22 developed markets increases uniformly in absolute value, from

−0.50% to −2.80% per month, as one moves from high institutional ownership (INST3) to high retail ownership portfolio (INST1). Thus, the monthly alpha difference between the long-short portfolio spreads using international stocks with the highest versus the lowest institutional ownership amounts to 2.31% per month. Similar results are obtained for stocks trading in G10 and G7 countries.

Overall, the results in Tables O.A.33–O.A.35 show that the low-risk anomaly exists only when the price impact of lottery demand falls disproportionately on high-IVOL stocks and is concentrated in stocks with high levels of retail ownership. Thus, we conclude that retail investors' demand for high-volatility stocks with large lottery payoffs induce them to pay high prices for these stocks, leading to temporary overpricing and negative future abnormal returns for such stocks.

5.4. Wealth inequality based explanation of the IVOL puzzle

Our household-level analyses have focused on the Swedish stock market and utilized the panel data on Swedish households. The choice of Sweden and the Swedish stock market as an empirical laboratory to investigate the relationship between rich individuals and the low-risk anomaly is motivated by several reasons. First and foremost, the unique household level data allow us to directly measure the holdings of the rich investors at the individual security level, which is not feasible in many other markets. Second, many aggregate economic and financial indicators of Sweden resemble those of the U.S. The measures of wealth inequality, GDP per capita, and stock market participation rates are highly similar, for instance. Thus, Sweden provides a unique setting to test the impact of (wealthy) individual investors on stock prices. At the same time, though, it may raise concerns that our findings are specific to Sweden. To provide external validity, we next present some indicative cross-country evidence.

Our findings, thus far, conform to the notion that the demand of rich households partly drives the IVOL puzzle. This suggests that, at the aggregate level, the negative return spread between high- and low-IVOL stocks should be higher when aggregate wealth is more unevenly distributed across the population, so that the bulk of wealth is concentrated in the hands of a small number of rich households. In other words, the IVOL puzzle should be stronger in countries with higher wealth inequality.

To test this hypothesis, we next investigate the relation between wealth inequality and the IVOL puzzle across countries. Specifically, we focus on a sample of developed countries, and first calculate the annual long-short return spreads on IVOL-sorted portfolios for each country in each year using individual stock data from the respective market for the time period between 1991 and 2017.³¹ Then,

we group these countries into two using the median share of wealth held by the top 10 percent of households in the wealth distribution across these countries, and contrast the average IVOL effect across the (relatively) high and low wealth inequality countries.

The univariate portfolio sorts indicate a negative relation between the IVOL effect and wealth inequality as measured by the top 10 wealth share.³² Specifically, we observe that the mean IVOL effect accounts for −9.46% per annum in countries with higher wealth concentration, while this figure accounts for −3.66% per annum in countries with relatively lower wealth concentration at the top. Importantly, the difference of −5.81% per annum is also significant with *t*-statistics ranging from −1.97 to −2.25, depending on the specification. Overall, the cross-country evidence provides some indicative support for the external validity of our findings from the Swedish data.

6. Conclusion

In this paper, we investigate the role of individual investors, particularly the rich individuals, in the existence of low risk anomalies. We begin our analysis by studying the stock investment behavior of individual investors at the household level with a particular focus on whether rich households tend to tilt their portfolios towards stocks with higher IVOL. Our high-quality household panel dataset enables us to identify the relative position of each household in the entire wealth distribution, and to directly observe and analyze their stock investment behavior over the 1999–2007 period.

Our household level analysis shows that increases in wealth tend to exhibit opposing effects on the risk-taking behavior of households who are in different social classes as measured by their relative position in the wealth distribution. Specifically, increases in household wealth have a significant positive effect on the portfolio IVOL of richer households but a negative effect for households in the bottom 90% of the wealth distribution. We also show that households who move up to the top of the wealth distribution (i.e., top 10, top 5, or top 1) significantly increase their stock portfolio exposure to IVOL. These results remain intact after accounting for a rich set of household and portfolio characteristics, and a battery of fixed effects.

Overall, the findings of the household level analysis are consistent with the model predictions of Roussanov (2010) who shows that the relative status concerns associated with wealth are more important for the rich households, and thus, the (status-seeking) rich optimally choose to invest in assets with highly idiosyn-

³¹ The choice of countries and time period in this analysis are dictated by the data availability for wealth inequality and stock information, respectively. We use the most recent data available for the top 10 shares of developed countries from the OECD Wealth Distribution Database. The information for Sweden comes from Lundberg and Waldenström (2018). We are not able to find wealth inequality data for Hong Kong, Israel, Singapore, and Switzerland, so we exclude these countries from this analysis.

³² For each sampled stock in the analysis, we calculate its idiosyncratic volatility as the standard deviation of the residuals from time-series regressions of the daily excess stock returns on the daily excess market returns and the daily size (SMB) and book-to-market (HML) factor returns in the past year. We require at least 200 daily return observations (in a year) to be available in order to calculate the IVOL. In each year, stocks trading on the respective stock markets are sorted into decile portfolios based on their IVOL. We then calculate the excess return on the arbitrage portfolio with a long position in the portfolio of highest IVOL stocks (decile 10) and a short position in the portfolio of lowest IVOL stocks (decile 1) for each country in each year.

cratic and positively skewed payoffs in order to advance their relative wealth position and get ahead of their peers. We next consider a number of explanations for our results in the process of exploring the economic underpinnings of the demand of rich households for stocks with higher idiosyncratic volatility. Our analysis reveals that social status concerns of rich households and the induced lottery preferences provide an explanation for their tendency to invest in high-volatility stocks. On the other hand, alternative explanations, including family-controlled firms and insider holdings, constraints on financial leverage, and co-skewness risk, receive no support from the data.

Building on these findings, we then investigate the implications of our findings for aggregate stock market outcomes, particularly for the existence of the IVOL puzzle in Sweden. We first show that stocks with high-IVOL generate low risk-adjusted returns, which is largely driven by the underperformance of high-volatility stocks. We further provide evidence that high-IVOL stocks are subject to significant overpricing, and the IVOL puzzle is mainly driven by individual investors' demand for high-IVOL stocks, which leads to temporary overpricing and negative future abnormal returns. To understand the potential role of rich individual investors in the existence of the IVOL puzzle, we then form bivariate portfolios of stocks sorted by IVOL and stock ownership levels of rich individual investors. We find that the anomalous negative relation between risk and future risk-adjusted returns is only confined to those stocks held by rich households, whereas there is no evidence of the low risk anomaly for stocks held by non-rich individual investors. We also verify these findings using a difference-in-difference type design where we focus on a subset of stocks. Specifically, we exploit exogenous variation in the aggregate demand of the rich households for individual stocks generated by a rule exception in inheritance and gift taxation of household stock wealth, and show that an increase in rich households' aggregate demand for a particular stock leads both to an increase in the IVOL of that stock and to significantly more negative risk-adjusted returns in the following month.

Finally, we propose a status-driven lottery demand factor and test if this newly proposed factor produces significant return and alpha spreads. We find that the new factor, constructed based on lottery demand and stock ownership levels of rich individual investors, produces economically large and statistically significant premium. More importantly, this newly proposed factor explains the idiosyncratic volatility puzzle in Sweden. Thus, we conclude that social status concerns of rich households and the induced lottery preferences explain wealthy investors' demand for high-risk stocks, leading to overpricing and low future returns for such stocks.

Declaration of Competing Interest

None

Data availability

Data will be made available on request.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jfineco.2023.103715](https://doi.org/10.1016/j.jfineco.2023.103715).

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